Beaver River Conservation and Management Plan

FINAL

August 2008

pennsylvania environmental council

DCNR

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EXECUTIVE SUMMARY

Purpose of the Plan

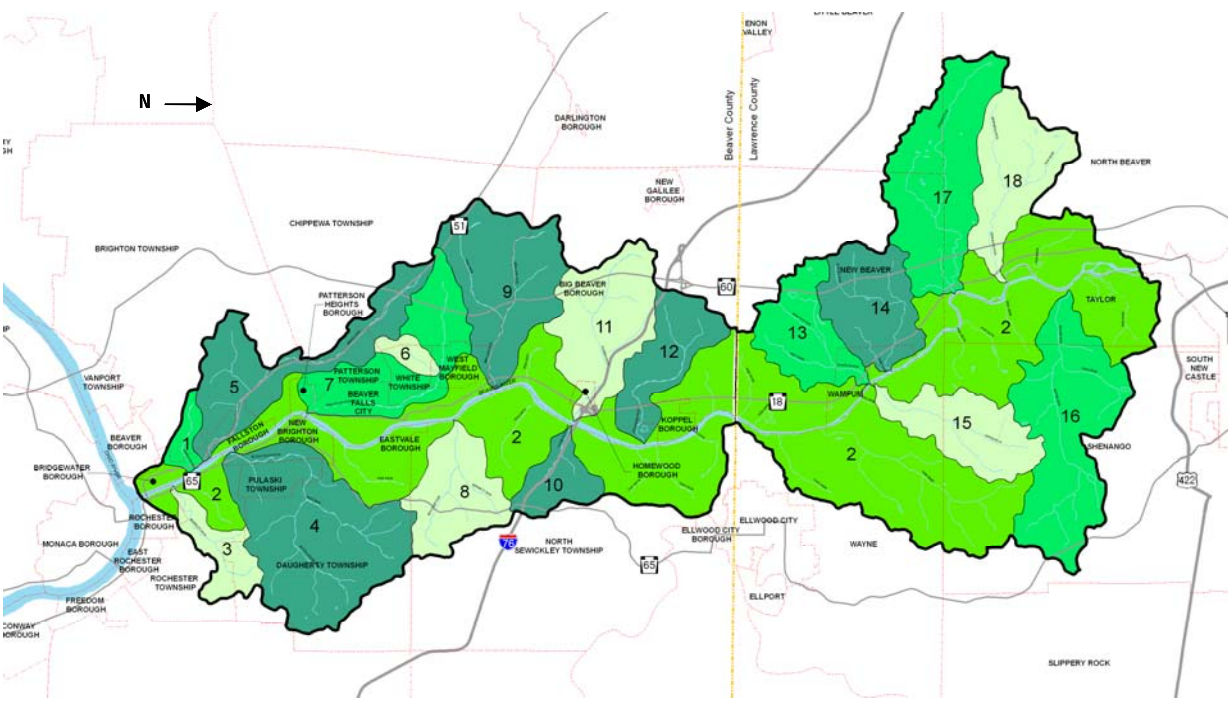
The Beaver River Conservation and Management Plan (Plan) presents mechanisms to promote natural resource conservation and enhancements in more than two dozen western Pennsylvania communities. The Plan addresses local and regional conditions and concerns while embracing state-wide conservation initiatives.

Assessments of the area's sensitive natural resources, land use and infrastructure patterns, historical flooding, and public policies form the basis of the Plan. From this information, a range of projects and policies are outlined for county, regional, local and other community stakeholders to implement. The Plan expands upon the fundamental conservation elements encouraged by the Pennsylvania Department of Conservation and Natural Resources (DCNR) as the foundation for identifying potential resource management recommendations.

The Beaver River Conservation and Management Plan was produced with financial assistance from DCNR's Community Conservation Partnership Program. Matching funds were provided by the Mellon Foundation and The Beaver County Community Development Program. The Plan was prepared by the Pennsylvania Environmental Council with analysis and planning completed by Environmental Planning and Design, LLC.

Overview of the Beaver River Watershed

The 22-mile Beaver River is formed by the confluence of the Mahoning and Shenango Rivers just south of New Castle, Lawrence County. The river flows south through Beaver County until it empties into the Ohio River.



From its beginnings, this Plan was intended to be innovative and far-reaching. This study's boundary reaches beyond a standard river setback prescribed in many similar types of studies in the state to include the full extent of 17 tributaries draining into the Beaver River. Being both land- and water-based, the Plan increases environmental stewardship of our natural assets while allowing for continued economic growth.

The entire Beaver River Watershed consists of 58,000 acres in a diverse mix of urban and rural lands. Urban communities dot the watershed's southern end, approaching the Beaver's confluence with the Ohio while more rural landscapes characterize the watershed's central and eastern sections.

The watershed is home to a wide range of natural resources, including waterways, steep slopes, biological diversity areas, and riparian areas. Many of these resources are considered to be "sensitive" based upon their rarity or location relative to development. Initial mapping of these sensitive resources found many to be in areas where population currently exists or could grow in the future.

Development of the Plan

An Advisory Committee comprised of representatives from local and county government, environmental and community organizations, and recreation interest groups had primary responsibility for development of the Plan. Working in conjunction with the Pennsylvania Environmental Council and Environmental Planning and Design LLC, the project manager and consultant, respectively, the Committee provided overall direction, reviewed draft analyses and recommendations, and acted as a liaison to the region's communities and interest groups.

Substantial amounts of public input were gathered throughout the creation of the Plan. Ideas and feedback from individuals and community groups were gathered through public and committee meetings and municipal surveys.

The Plan's overall goals emerged from this process. The goals include:

- Identify locations of sensitive resources in the Beaver River Watershed;
- Catalog the extent of existing water and sanitary sewer service infrastructure in environmentally sensitive areas;
- Evaluate the relationship between existing communities' development and potential build out patterns;
- Assess potential flood implications (normal and severe conditions) based on future development;
- Present opportunities for public education and awareness about conservation; and
- Provide information to aid in municipal and regional decision making and planning.

The Advisory Committee, project manager, and consultants used these goals, together with data collected in the first part of the process to identify challenges facing the Beaver River Watershed and opportunities for action.

Key Issues within the Watershed

After collecting pertinent data, the Advisory Committee and community stakeholders analyzed the information and suggested a set of issues and opportunities to be considered. The key issues that the counties and communities explored were:

- The extent, significance, and relationship of environmental, historical, and cultural resources;
- Flood prone areas;
- Expansion of public infrastructure into naturally sensitive areas; and
- Potential growth and changes permitted by existing zoning affecting existing sensitive resources.

The relationship between environmentally significant habitats and potential recreational opportunities are highlighted on the Conservation Strategy Analysis Map, which appears after the following section.

Key Recommendations and Actions

Public comments were integrated into the Plan's technical analyses to develop a series of recommendations and actions. These recommendations are related to public policies, programs, and activities. Multiple stakeholders, including government agencies, non-profit conservation/ community organizations, educational/cultural institutions, business interests and property owners, will be involved in implementation. The Plan also advocates that the watershed's counties and municipalities adopt resolutions that endorse the general and specific recommendations set forth for implementation.

The following recommendations and actions are considered to be those of highest priority. Additional projects, policies, stakeholders, and timeframes are outlined in the Action Plan.

1. *Identify or form a conservation and management advisory body to implement the recommendations of the Conservation and Management Plan*

   The watershed's communities should identify or form a conservation and management advisory body that would assist in identifying on-going challenges to conserving the region's natural resources, promote environmental programming, lead the preparation of ordinances and detailed inventories, make recommendations for the use of available open spaces, and advise government agencies on conservation issues. One possible example is the creation of an Environmental Advisory Council.

2. *Create a River Corridor Management Ordinance*

   Representatives of the two counties should work together with local municipalities to develop and adopt a River Corridor Management Ordinance as part of their Act 167 Plans. This Ordinance would address potential development, implementation, and enforcement of programs related to management of stormwater from new development and redevelopment.

3. *Adopt impervious surface provisions as part of the River Corridor Management Ordinance to minimize negative impacts of stormwater run-off*

Communities within the Beaver River Study Area should coordinate to adopt impervious surface provisions. Impervious surface provisions can be adopted on either a local or watershed-wide basis.

4. *Advocate for the creation and adoption of a Steep Slope Ordinance*

   Currently, several communities within the watershed limit the amount of disturbance permitted to occur on steep slopes. Based on the Plan's findings, the ordinance provisions should be adopted to limit the amount of disturbance that should occur on slopes greater than or equal to 25 percent and prohibit any additional development on slopes greater than or equal to 40 percent.

5. *Evaluate the feasibility of other watershed-wide ordinance updates that coordinate development and infrastructure opportunities*

   Beaver and Lawrence Counties should evaluate the feasibility of potential ordinance language and/or mapping updates which encourage the coordination of potential development and infrastructure.

6. *Promote development that respects the landscape's character and capacity*

   Counties and communities should encourage development patterns in urban, suburban, and rural areas of the Beaver River Watershed that are compatible with existing character and infrastructure service while minimizing negative impacts to sensitive natural resources.

7. *Establish consistent review procedures for identification of conservation need*

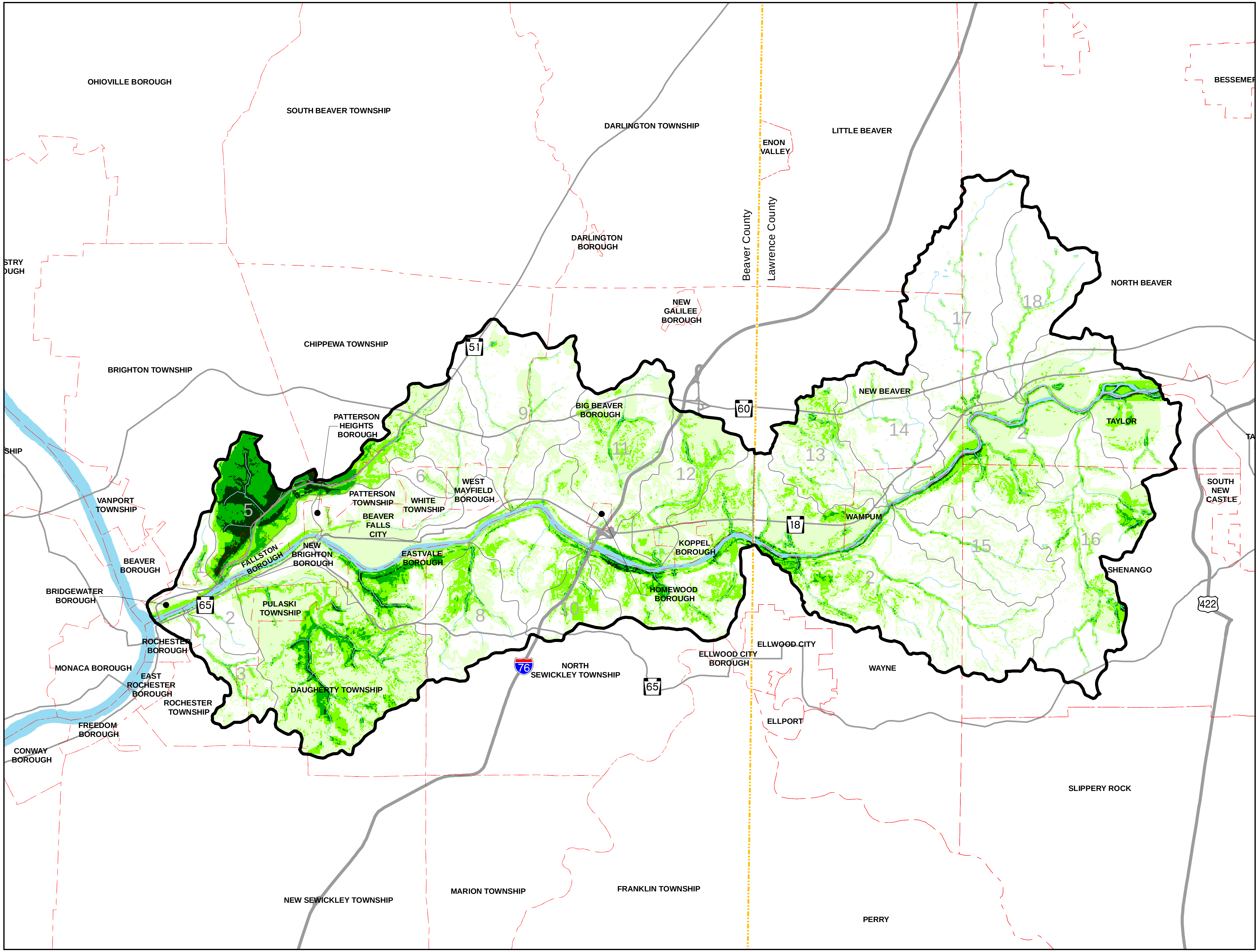
   County and regional stakeholders should work with local municipalities and the development community throughout the Study Area to establish a formal, consistent review process to identify how existing natural resources could be conserved.

8. *Coordinate conservation with other county open space and greenway efforts*

   The counties and identified stakeholders should continue to work to coordinate how projects, policies and funding opportunities can be optimized and implemented succinctly.

9. *Initiate discussions with local land trusts and conservancies to conserve lands located in the River Corridor*

   The leaders and residents of Beaver and Lawrence Counties should initiate discussions with local land trusts and conservancies to identify how existing scenic, recreational, education and environmental landscapes within the watershed could offer opportunities for land conservation and promote a healthier watershed environment.



Legend

- Study Area Boundary
- Watershed Boundary
- County Boundary
- Municipal Boundaries
- Highways
- Major Roads
- Rivers/Streams/Lakes

TOTAL SCORE

- 1 - 5
- 6 - 10
- 11 - 15
- 16 - 28

Conservation Matrix

		Resource	Weighted Value
Environmentally Sensitive Lands	a	Exceptional/High Quality Watersheds (Presence of Exceptional/High Quality Streams. Includes streams with trout stock potential.)	5
	b	Wetlands	5
	c	Steep Slopes (>25%)	4
	d	Stormwater Filtration	4
	e	Floodplains	2
	f	Groundwater Recharge	1
Habitat	a	BDAs	5
	b	Habitable Woodlands	5
	c	Warmwater Gamefish Habitats	3
	d	Important Bird Areas	3
Recreation	a	Trails	3
	b	Non-motorized Boating/Fishing Access (combines flatwater, whitewater and sail boating)	3
	c	Primitive Camping	2
	d	Hunting	2
	e	Motorized Boating/Fishing Access	1
	f	Non-Primitive Camping	1
	g	ATV areas	0

Sub-Watersheds

1. Hamilton Run
2. Beaver River
3. McKinley Run
4. Block House Run
5. Brady Run
6. Grimms Run
7. Walnut Bottom Run
8. Bennett Run
9. Wallace Run
10. Thompson Run
11. Clarks Run
12. Stockman Run
13. Wampum Run
14. Eckles Run
15. Snake Run
16. McKee Run
17. Jenkins Run
18. Edwards Run

1,000 Acres
500 Acres
250 Acres

CONSERVATION STRATEGY ANALYSIS

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: September 2006
1989.06.25r1

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NORTH

ACTION PLAN		
Recommendations	**Participant Groups**	**Project Initiation Timeline**
1. Management		
A. Identify or form a conservation and management advisory body to implement recommendations of the Conservation and Management Plan (referred to in the recommendations as "river advisory body").	County Governments, municipalities	Immediate
B. Encourage communities to participate in the Act 167 Process	County Governments, municipalities, river advisory body	Immediate
C. Create a Watershed Management Ordinance that promotes watershed conservation principles and techniques	County Governments, municipalities	High Priority
D. Preserve the scenic quality of the Beaver River Corridor including biological diversity areas, natural heritage areas, steep wooded slopes, riparian (streamside) areas, wetlands, islands and other similar characteristics	County Governments, municipalities, conservation organizations, river advisory body	High Priority
E. Preserve or enhance contiguous forests in identified sensitive landscapes by developing a Forest Management Plan for the region	County Governments, municipalities, land owners, Bureau of Forestry, conservation organizations, river advisory body	High
F. Conduct an Agricultural Practices Assessment to identify appropriate farmland conservation and runoff management opportunities	Conservation Districts, County Governments, river advisory body	Long Term
G. Clean up old dock posts from washed out docks	Dock owners, municipalities	Medium Priority
H. Develop, in cooperation with stakeholders, an annual report updating key indicators of watershed health and describe recent progress in preserving and enhancing watersheds, new findings, and study results	River advisory body, Conservation Districts	On-going
I. Gauge and build public support for water conservation and recycling	County Conservation Districts, County Governments, river advisory body, conservation organizations, schools	Medium Priority
J. Conduct a thorough plant and wildlife inventory	Conservation organizations, educational institutions	Medium Priority

K. Clean up illegal dumpsites and litter along the River	Municipalities, conservation or environmental organizations	On-going
L. Study the water quality of the Beaver River watershed	PA DEP, Conservation Districts	High Priority
2. Development and Infrastructure		
A. Encourage redevelopment and in-fill of brownfields (previously developed sites)	County Governments, Municipalities, Development Organizations/Corporations, Authorities, Utilities	On-Going
B. Promote development within existing infrastructure service areas	County Governments, Municipalities, Development Organizations/ Corporations, Authorities, Utilities, river advisory body	On-Going
C. Align new development and future infrastructure expansion to minimize impacts on sensitive natural resource areas	County Governments, Municipalities, Development Organizations/Corporations, Authorities, Utilities	High-Priority
D. Build collaboration between the public and private sectors to update aging and/or under-serviced stormwater management systems	County Governments, Municipalities, Development Organizations/Corporations, Authorities, Utilities, Conservation Districts, river advisory body	On-Going
E. Consider repair and reestablishment of riparian (streamside) vegetation in impacted watersheds	Conservation Districts, river advisory body	Medium Priority
F. Sponsor septic system education for residents of unsewered portions of the Beaver River Corridor	PA DEP, Conservation Districts, river advisory body	High Priority
G. Reduce accumulation of sediment in infrastructure systems by addressing point and nonpoint pollution sources	County Governments, Municipalities, Conservation Districts, river advisory body, PADEP	Medium Priority
H. Address stormwater issues and sedimentation problems within the Beaver River corridor in context of existing and future Act 167 and MS4 plan recommendations	County Governments, Municipalities, PA DEP, Conservation Districts, river advisory body	Medium Priority
I. Remove invasive plant species in public riparian (streamside) areas and in private lands where possible. Continue to monitor riverfront lands for invasive species	Conservation Districts, river advisory body, conservation organizations	On-going
J. Promote planting of native species in newly developed riverfront areas	Municipalities, Conservation Districts, river advisory body, conservation organizations	High Priority

K. Promote use of pervious pavements, gravel access roads, and other means to encourage stormwater infiltration into the water table	Conservation Districts, Municipalities, river advisory body	Medium Priority
L. Implement riverfront development strategies identified in other studies including the area's Riverfront Development Plan	County Governments, Municipalities, Development Organizations/Corporations	Medium Priority
3. Outreach, Education, and Public Awareness		
A. Pursue funding to support regional-oriented staff for the river advisory body in its coordination and implementation of this Plan's recommendations	County Governments	High Priority
B. Create a Beaver River Watershed Website related to the projects outlined in the Conservation and Management Plan	County Governments, river advisory body, Conservation Districts	Medium Priority
C. Coordinate conservation and management initiatives with local schools and institutions of higher education to promote research/implementation opportunities including development of primary and secondary school education curricula related to the natural resources of the Beaver River Watershed	Conservation Districts, local non-profits, schools, river advisory body	Medium Priority
D. Work with the railroads, municipalities, and the PA Fish and Boat Commission on designating safe active and passive river access locations	Trail Groups, local non profits, Conservation Districts, PA Fish and Boat Commission, County Governments, Municipalities, river advisory body, railroads	Medium Priority
E. Establish scenic driving tours of the River valley, including stops for train viewing, scenic river views, and historic/cultural attractions	Conservation Districts, local non-profits, historical societies, river advisory body	High Priority
F. Help to coordinate input to, and distribution of, outreach newspapers published by agencies and community groups	Conservation Districts, local non-profits, river advisory body	Medium Priority
G. Encourage and assist local and regional agencies to incorporate interpretive and educational features as part of recreational facilities and other public works projects	Conservation Districts, Municipalities, river advisory body, local non-profits	On-going

H. Promote active stewardship among commercial and residential stakeholders; bring this message/updates to advisory boards, environmental commissions, planning commissions, and other venues for public input to agency decision making	Conservation Districts, local non-profits, conservation organizations, river advisory body	High Priority
I. Sponsor semi-annual hazard and toxic disposal collection programs	Conservation Districts, local non-profits	Medium Priority
J. Establish an Adopt-A-Creek program	Conservation Districts, local non-profits	Long Term

Next Steps

Upon DCNR's and local municipal approval, the Beaver River Conservation and Management Plan will be submitted for inclusion on the Pennsylvania Rivers Registry, qualifying it for DCNR matching grants to municipalities and nonprofits for implementing the recommendations provided in this report. Communities are encouraged to update the Action Plan regularly to chart progress on achieving recommendations.

If you are interested in implementing projects listed in the Action Plan, here are several steps that may help you.

1.) Contact other potential partners, including local municipal officials, to find out if there is an interest in collaborating on the project. You do not have to be listed as a potential partner or stakeholder to initiate or implement a project. There are probably many other people who are interested in working on a project than are listed. Also, make sure that no one else is working on the same or similar project.

2.) Brainstorm with other partners on how you might implement the project. Develop goals, objectives, and a list of tasks that will take you to your desired outcome.

3.) Determine your funding needs and research potential funding sources. DCNR provides grants for planning and technical assistance, development, and acquisition. The Department of Environmental Protection (DEP) offers grants that address non-point source pollution. If you are thinking about applying for funding from one of these agencies, please contact the local agency representative to discuss the potential for your project as soon as possible. Many other funding sources are available. The Pennsylvania Organization for Watersheds and Rivers (POWR) has several options listed on their website (www.pawatersheds.org).

4.) Prepare and submit grant applications, paying careful attention to the grant requirements and deadlines.

5.) If a grant is awarded, convene local partners and notify the media.

Left: View of the Beaver River from Geneva College.

Cover: View of the Beaver River below Townsend Dam.

Copies of the entire Beaver River Conservation and Management Plan can be found on the Pennsylvania Environmental Council's website: www.pecpa.org.

pennsylvania environmental council

This project was completed by the Pennsylvania Environmental Council (PEC). PEC protects and restores the natural and built environments through innovation, collaboration, education, and advocacy. Comments or questions about this Plan can be directed to PEC at 412-481-9400 or 22 Terminal Way, Pittsburgh, PA 15219.

DCNR

This project was financed in part by a grant from the Community Conservation Partnership Program, Keystone Recreation, Park and Conservation Fund, under the administration of the Pennsylvania Department of Conservation and Natural Resources, Bureau of Recreation and Conservation. For more information about DCNR's programs, visit their website: www.dcnr.state.pa.us.

Matching funds for this project were provided by the Richard King Mellon Foundation and the Beaver County Community Development Program.

CHAPTER 1: INTRODUCTION

The Pennsylvania Department of Conservation and Natural Resources' (DCNR) Rivers Conservation Program was developed to conserve and enhance river resources through preparation and accomplishment of locally-initiated plans. All plans contain a summary of available information about the water resource, which makes the document a useful educational tool. However, the substantive sections of these plans are the recommendations developed through the analysis and the action steps that will move the recommendations forward. The plans are an important tool for local governments, organizations, and businesses to aid in decision making and municipal planning.

The Beaver River was selected for study because of its potential as a recreational amenity, the current development pressures in the watershed, and a lack of a significant watershed group in the area with which to address conservation issues and opportunities.

About the Authors

The Plan was completed by the Pennsylvania Environmental Council (PEC), a statewide, non-profit organization devoted to promoting the protection of watersheds, the sustainable uses of land, and the implementation of environmental innovations. Since its founding in 1970, PEC has been centrally involved in a broad range of issues affecting the region's environmental recovery and ever-improving quality of life. More information about PEC can be found at www.pecpa.org.

The Plan was prepared by PEC staff member Janette M. Novak and former staff member Beth Brennan Fisher with analysis and planning completed by Environmental Planning and Design, LLC.

Support and Funding

DCNR's Community Conservation Partnership Program, administered by the Bureau of Recreation and Conservation, awarded a Keystone, Recreation, Park and Conservation Fund grant to PEC in 2004 to complete the Beaver River Conservation and Management Plan. Matching funds were provided by the Richard King Mellon Foundation and the Beaver County Community Development Program.

The next step in the planning process is to obtain local support to seek admission of the Plan for inclusion on the Pennsylvania Rivers Registry. This registry was created to promote river conservation and recognize rivers or river segments in communities that have completed River Conservation Plans. Once registry status is achieved, municipalities and non-profits are able to apply for DCNR implementation, development, or acquisition grants based on the Plan's recommendations.

Visit www.dcnr.state.pa.us for more information about DCNR grants and to view maps and Conservation Plans from around the state.

Frequently Asked Questions about River Conservation Plans

Some communities and residents who are not familiar with the Rivers Conservation Program and the planning process have concerns about the implications of having a Rivers Conservation Plan. Following are three common questions about the program.

1. How will this project impact private property?

The Plan itself will not encroach upon private property and will not dictate what landowners can and cannot do. The Plan is an opportunity to work towards a vision for the future of the River; it is not focused on specific property owners. Recommendations may be made within the plan for general care of lands adjacent to waterways; however, it will be up to local landowners and municipalities if they want to follow these recommendations or not. Property owners are encouraged to participate in terms of developing the greater vision for the area.

2. Are you proposing land use restrictions in the viewshed of the river corridor?

A Plan may encourage updating county and municipal plans and ordinances to include things like natural resource protection zoning. Such zoning often affects new development only and such actions would then need to be adopted by county and local governments. The Plan in itself cannot require land use restrictions – only local governments can carry out such actions. If local governments pursue such zoning amendments or changes, it is required by the Municipalities Planning Code to be a public process.

3. Can the Plan enable new environmental regulations that could affect riverfront property owners or river-side industry?

The Plan itself cannot enable or force any government agency to enact new legislation. What it can do is provide information and public comments about the Beaver River that could be used as a guide in creating or amending legislation. Those types of actions generally would require a public process.

Acknowledgements

An Advisory Committee, comprised of government representatives, environmental and community organizations, and recreational interest groups identified data sources, reviewed draft Plans, and acted as liaisons to and for their community or interest group. Public input was gathered through public meetings, key person interviews, stakeholder meetings, and municipal surveys.

PEC gratefully acknowledges the following people who served on the advisory committee or contributed to the development of the Plan:

Charles Camp, Steve Craig, Matthew Cucinelli, Alan DeSanzo, Dan Donatella, Jack Erath, Bill Evans, Jim Gagliano, Megan Gahring, Tom Hamilton, Donald Inman, Helen E.

Jackson, Thomas Ketterer, Frank Mancini, John Martino, Kathryn McClure, Amy McKinney, Shirley Mesing, Larry Morley, Pashek Associates, Ed Piroli, Greg Powell, Jeanne Rarick, Laura Rubino, Tom Scuoteguazza, Joe Spanik, Todd Stevenson, Lisa Troiani, Carolyn Verszyla, Donald Wachter, Dan Woodske, Doniele Andrus.

Supplemental Information

Additional information can be found in the following documents, which are valuable resources and can be considered supplemental resources for this Plan.

- Beaver County Comprehensive Plan (1999), Gannet Fleming
- Beaver County Comprehensive Recreation and Parks Plan (2003), Pashek Associates
- Beaver County Greenways and Trails Plan (2007), Pashek Associates
- Inventory and Assessment of Historic and Heritage Sites in Beaver County (1998)
- Beaver County Riverfront Development Plan (1993), Beaver County Corporation for Economic Development
- Bridging the Confluence Communities: Building a Shared Vision for Riverfront Development (2005), Beaver Initiative for Growth, Beaver County Corporation for Economic Development
- Natural Heritage Inventory for Beaver County (1993), Western Pennsylvania Conservancy
- Rivers of Destiny (1999)
- Natural Heritage for Lawrence County (2002), Western Pennsylvania Conservancy
- Lawrence County Comprehensive Plan (2004), Lawrence County Planning Commission
- Lawrence County Recreation Study (c. 1975), Lawrence County Planning Commission
- Natural Infrastructure (2005), Southwest Pennsylvania Commission, The Heinz Endowments, Pennsylvania Department of Conservation and Natural Resources, Pennsylvania Environmental Council, Environmental Planning & Design
- Lawrence County Greenways and Open Space Plan (2008), Lawrence County Planning Department
- North Central Beaver County Southern Lawrence County Regional Economic Development Study

CHAPTER 2: CONSERVATION STRATEGY

The Beaver River Conservation and Management Plan (Plan) seeks to outline a realistic and effective approach for implementing future conservation-oriented efforts for 58,000 acres in Southwestern Pennsylvania. Similar to the format of many River Conservation and Management Plans that Pennsylvania communities are developing, this Plan inventories the various ecological and historical factors that have influenced the formation of the landscape surrounding the Beaver River.

As development patterns change, policy decisions within a community or a region can influence a watershed's characteristics. Consequently, this study of the Beaver River and its surrounding lands enhances the typical components of a River Conservation and Management Plan. This Plan:

1. Evaluates the potential development impacts of existing community zoning ordinances;
2. Assesses the relationship of historical flooding patterns and development patterns; and
3. Outlines a broad range of project and policy recommendations for county, regional, and local governing bodies as well as other community stakeholders to implement and effectively manage future conservation efforts.

The key components of developing a conservation strategy for this area include defining the study area, completing mapping analyses, and evaluating opportunities and challenges. These activities have shaped the ways in which strategy alternatives were evaluated. From these components, the Plan's series of project and policy recommendations for implementation have been created.

General Planning Process

The Beaver River Conservation and Management Plan presents a series of recommended policies and projects aimed to optimize the impacts of future conservation efforts on the lands adjacent to the Beaver River. The overall approach of analyzing conservation opportunities and challenges stems from an evaluation of existing and planned land use patterns, river-related natural resource inventory data, historic flooding activity, development ordinance provisions, infrastructure patterns and aerial photography. As mentioned above, the primary steps in preparing the overall conservation strategy include: defining the study area; completing inventory and analysis; and evaluating opportunities and challenges.

Defining the Study Area

The Plan encompasses more than 58,000 acres of Beaver and Lawrence Counties. In addition to the two counties, twenty-seven (27) municipalities have jurisdiction within the land studied as part of this Plan.

The boundary of a watershed is based upon topography with ridge tops and highpoints in the landscape forming the watershed's perimeter. Typically, as found throughout most of Southwestern Pennsylvania, municipal lines and watershed boundaries do not coincide with one another.

For the purpose of this Plan, the areas studied have been defined as the Study Area Watershed, Principal Watershed, and Tributary Watershed. This Plan's Study Area Watershed refers to the overall extent of land to which the proposed conservation and management strategies apply. The Study Area Watershed begins at the confluence of the Shenango and Mahoning Rivers – the two rivers which meet to form the body of water known as the Beaver River. The southern end of the

Study Area is where the Beaver River meets the Ohio River. The Study Area also includes land on both sides of its banks until reaching the nearest highest points in the landscape. The Beaver River flows generally through the middle of the Study Area.

The two sub-sets of the Study Area are its Principal Watershed and its Tributary Watersheds. Based upon topography, there is land along the river's banks where water drains directly into the river. This portion of the Study Area is known as the Principal Watershed.

Other portions of land in the Study Area drain into one of the river's tributaries, or "feeder" streams. There are 17 tributaries that flow into the Beaver River. The land surrounding each of the individual tributaries is known as a Tributary Watershed. There is one tributary of the Beaver River which is not included in the Plan - Connoquenessing Creek. Connoquenessing Creek flows into the Beaver River from the east and drains lands within Beaver, Lawrence, Allegheny and Butler Counties. A Watershed Conservation Plan was completed by the Western Pennsylvania Conservancy in February 2008. The entire plan, including its recommendations for that watershed, can be found at www.paconserve.org.

Diagrams illustrating the general relationship of the Study Area Watershed, Principal Watershed and Tributary Watersheds follow.

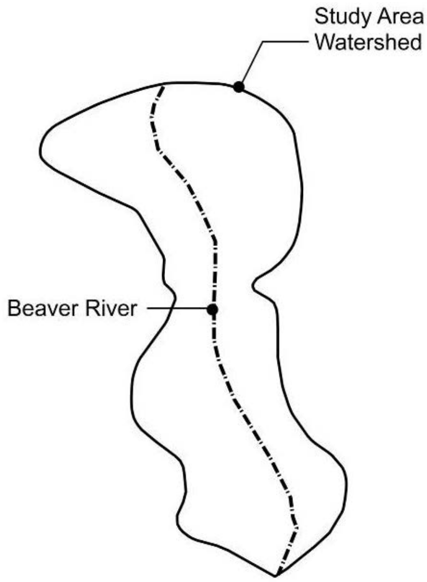


Figure 1: The Study Area Watershed represents all land included within the Conservation and Management Plan. The Beaver River is located in the middle of the Study Area.

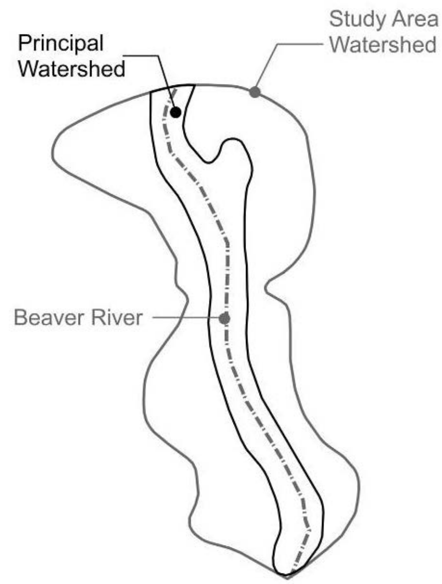


Figure 2: Water that flows directly into the Beaver River is within the Principal Watershed.

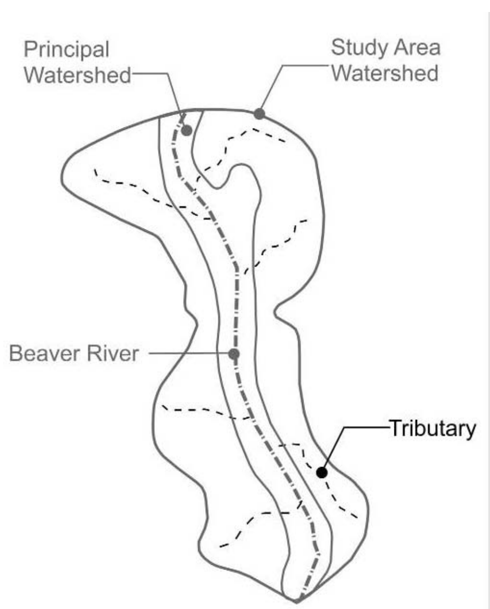


Figure 3: Tributaries of the Beaver River bring additional water into the Principal Watershed.

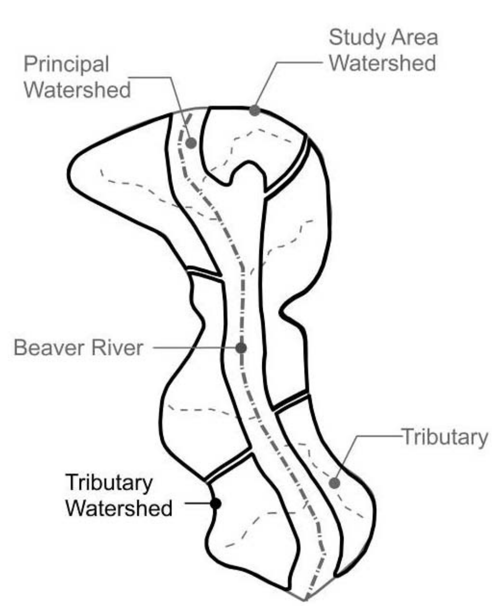


Figure 4: Land that slopes toward each tributary is known as a Tributary Watershed.

Description of Watersheds

The Study Area Map, following this section, identifies the specific locations and relationship of the River's Principal Watershed and its Tributary Watersheds. The Study Area's 18 watersheds include:

1.) Hamilton Run

Located primarily in Brighton Township and running through Bridgewater, the 363-acre Hamilton Run tributary watershed is one of the project's smallest drainage areas in the southern portion of the Study Area. Although there are areas of notable forest cover, a majority of the watershed is covered by low and high density urban land. Approximately half of the watershed is serviced with sewer and water infrastructure.

2.) Beaver River

In contrast to Hamilton Run, the Beaver River watershed is one of the Conservation and Management Plan's largest in geographic size, encompassing more than 21,000 acres and all communities bordering the waterway. A majority of land in the southern portion of this Principal Watershed, which does not possess steep slopes, is developed; more sparsely developed areas exist in the north. Communities situated along the Beaver River include Rochester Township, Rochester Borough, Bridgewater, Fallston, New Brighton Borough, Eastville Borough, North Sewickley Township, Big Beaver, West Mayfield, Koppel Borough, Wayne, Shenango, Taylor, North Beaver, Wampum, New Beaver and Homewood Borough. Infrastructure service along the River is generally concentrated to urbanized lands. Due to topography and historical development patterns, there are significant portions of this watershed that remain without sewer and water service.

3.) McKinley Run

Flowing through portions of Daugherty and Rochester Townships and the northern portion of Rochester Borough, the McKinley Run watershed is located at the southern portion of the Study Area near the confluence of the Beaver at the Ohio River. At nearly 930 acres in size, the McKinley Run watershed contains both extensive areas of developed land as well as landscapes supporting sensitive natural resources. Sewer and water service is generally limited to the watershed's western half.

4.) Block House Run

Block House Run drains stormwater and other run-off from land within portions of Daugherty Township, Pulaski Township and New Brighton Borough. Based upon existing land use patterns, areas of both developed and undeveloped landscapes exist in this 4,700-acre watershed. The presence of good warm water fish habitat is a notable characteristic of this watershed. Similar to McKinley Run, infrastructure service is generally accessible to development in the western portion of the Block House Run watershed.

5.) Brady's Run

This 2,700-acre watershed encompasses the eastern portions of Brighton and Chippewa Townships as well as the western sections of Patterson Township and Fallston Borough. The Brady's Run watershed is distinguished by the presence of a biological diversity area and its designation as an exceptional/high quality watershed. It is recognized in previous and current planning efforts, including the Beaver County Greenways Plan and the Western Pennsylvania Conservancy's Natural Heritage Inventory. Water and sewer service is generally not available within this watershed.

6.) Grimm's Run

Portions of White Township, Patterson Township, West Mayfield Borough, and Chippewa Township are situated within the 315-acre watershed. Infrastructure service is accessible to most land. Steep slopes are a common characteristic of the Grimm's Run watershed.

7.) Walnut Bottom Run

This 2,390-acre watershed encompasses Patterson Township, Beaver Falls, and White Township. Infrastructure service is accessible to most land surrounding Walnut Bottom Run. Similar to Grimm's Run and other smaller drainage areas, steep slopes are found in various portions of this watershed.

8.) Bennett Run

Primarily within North Sewickley Township and a small portion of Daugherty Township, the Bennett Run watershed contains 1,800 acres of urban land with existing infrastructure service as well as non-urban land (sparsely developed and/or undeveloped land). These non-urban lands are notable for the presence of sensitive environmental resources.

9.) Wallace Run

Two communities comprise the majority of the 3,060-acre Wallace Run watershed: Big Beaver and Chippewa Township. The northernmost portion of West Mayfield Borough is also included where Wallace Run flows into the Beaver River. Generally characterized by flatter topography than the rest of the Study Area this watershed contains prime warm water game fish habitat. Based on regional infrastructure mapping, approximately two-thirds of the land in the Wallace Run watershed has access to sewer and/or water service.

10.) Thompson Run

Located entirely in the mid-western section of North Sewickley Township, Thompson Run is the Study Area's third smallest watershed. The 891-acre drainage area generally parallels the Pennsylvania Turnpike (Interstate 76). With little existing development and no known access to water and sewer service, numerous natural resources including habitable woodlands and steep slope areas, are present.

11.) Clarks Run

Before intersecting with the Beaver River, Clarks Run flows through the mid-eastern portion of Big Beaver Township as well as a small section of Homewood Borough. There is a minimal amount of existing development and a considerable presence of natural resources within this 2,571-acre watershed

12.) Stockman Run

The 1,665-acre Stockman Run watershed is located in the northeastern section of Big Beaver and also encompasses a portion of Koppel Borough. Development currently within this area is minimal and of general low intensity throughout the drainage area. Infrastructure service is primarily concentrated to the easternmost portion of the watershed.

13.) Wampum Run

The Wampum Run watershed encompasses the southeastern section of New Beaver Township and the central part of Wampum, where Wampum Run meets the Beaver River. Nearly one-half of this 1,719-acre watershed is characterized as forested land. Based on the analysis of aerial photography, the remaining portion of the watershed possesses lower intensity residential and agricultural land coverages.

14.) Eckles Run

This 2,021-acre watershed lies primarily in the northeastern section of New Beaver and also in the southern end of Wampum. The general boundary of infrastructure service mirrors the limits of existing development. Development can be characterized as low intensity urban land coverage and woodlands.

15.) Snake Run

Encompassing lands both in Wayne and Shenango Townships, the 1,900-acre Snake Run watershed contains a limited amount of developed residential and non-residential land. The predominate land coverage of the watershed is classified as woodlands.

16.) McKee Run

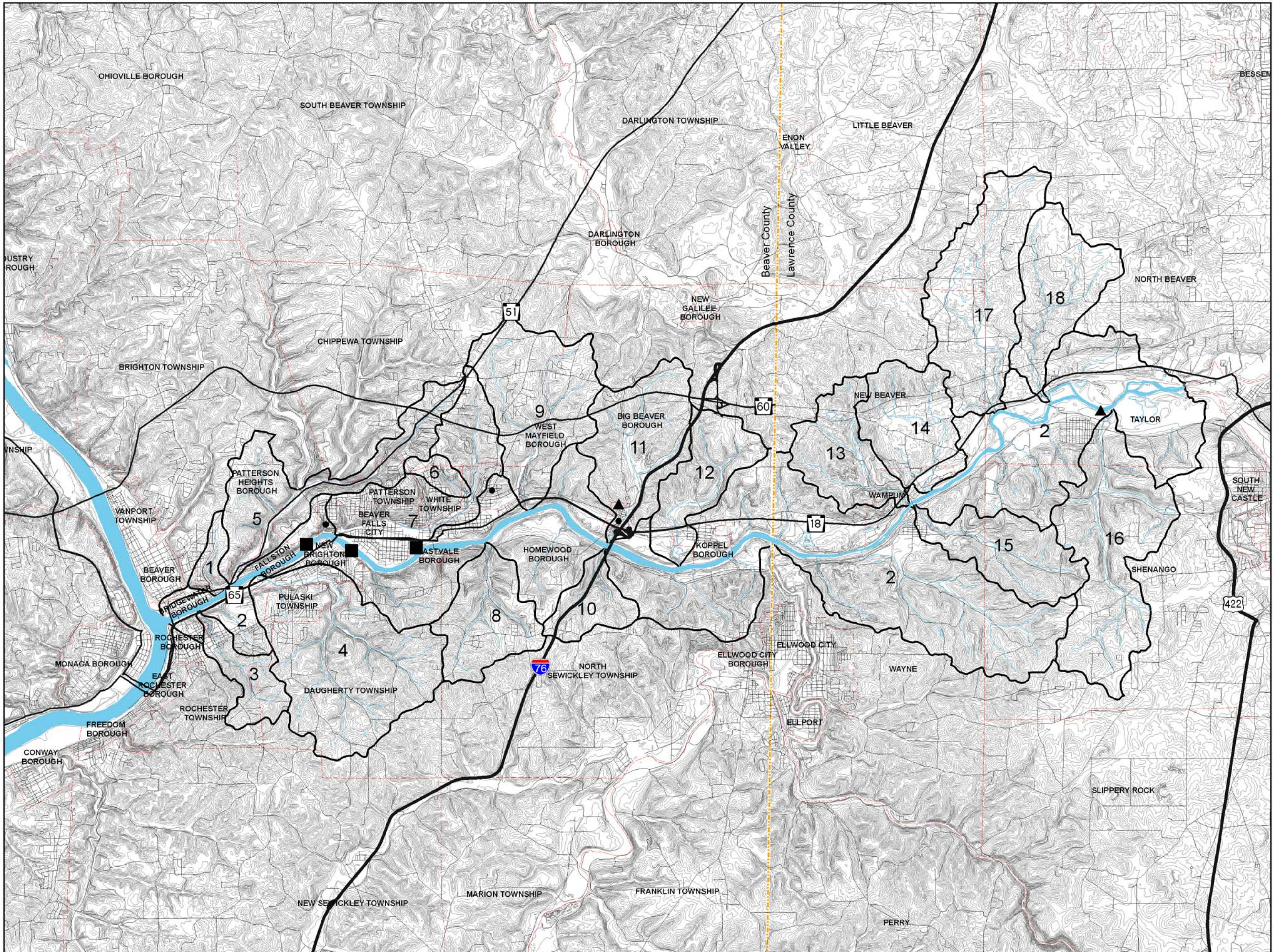
This watershed covers a small section of central and eastern Taylor Township with a majority of its land in Shenango Township. Most of the 3,965-acre McKee Run watershed is forested and agricultural land cover; land utilized for residential and non-residential development is both limited and scattered throughout the watershed. Water infrastructure, along with a limited amount of sewer service, can be found in the western portion of the watershed.

17.) Jenkins Run

The Jenkins Run tributary watershed lies in portions of North Beaver, Little Beaver and New Beaver Townships. This 3,590-acre watershed is almost entirely undeveloped characterized by the presence of agricultural and open space.

18.) Edwards Run

The Edwards Run tributary watershed lies in the southcentral portion of North Beaver Township. A limited amount of development exists within these 2,215 rural acres.



Legend

- Watershed Boundary
- County Boundary
- Rivers/Streams/Lakes
- Highways
- Major Roads
- Minor Roads
- Existing 20' Contours
- Hydroelectric Dams
- Other Dams

Sub-watersheds

1. Hamilton Run
2. Beaver River
3. McKinley Run
4. Block House Run
5. Brady Run
6. Grimms Run
7. Walnut Bottom Run
8. Bennett Run
9. Wallace Run
10. Thompson Run
11. Clarks Run
12. Stockman Run
13. Wampum Run
14. Eckles Run
15. Snake Run
16. McKee Run
17. Jenkins Run
18. Edwards Run

1,000 Acres

500 Acres

250 Acres

STUDY AREA MAP

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvania Environmental Council

Prepared by: Environmental Planning and Design, LLC

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NORTH

Completing Inventory and Analysis

As part of understanding the physical composition of each watershed, a series of detailed inventories and analyses were completed. These mapping and data efforts provide the foundation for establishing the Study Area's opportunities and challenges. The Study Area's resources were mapped using computer-based Geographic Information System (GIS) data available through the Southwestern Pennsylvania Commission's (SPC) Natural Infrastructure of Southwestern Pennsylvania database, Beaver County, Lawrence County, and other government-based agencies.

The full range of inventory and analyses completed as part of this Conservation and Management Plan is presented in detail in Chapter 4 of this Plan. In general:

- One series of analyses evaluated the presence of man-made patterns in each of the communities within the Corridor. Infrastructure, land coverage, and population density are examples of these patterns.
- Another series of analyses examined natural resource patterns. These analyses included those related to slopes, riparian (streamside) buffers, and sensitive habitats. Sensitive habitats identified include features such as floodplains, biological diversity areas, and general watershed quality.

Evaluating Opportunities and Challenges

From the completed analyses, a series of planning and policy opportunities and challenges emerge. These opportunities and challenges are influenced by patterns of natural resources and existing development, the potential impact of future growth as permitted by the communities' zoning ordinance provisions, and past flooding occurrences.

To optimize the positives that opportunities offer and minimize the negatives that challenges present, it will be necessary for county, regional and local entities to maintain on-going coordination of planning and policy efforts.

Analyzing Conservation Issues

The effectiveness of proposed conservation strategies is dependant on a comprehensive understanding of the historical and current landscape as well as local and regional planning and policy efforts. The Beaver River Conservation and Management Plan's strategies build upon several components including:

- Determining the presence and sensitivity of natural resources,
- Assessing impacts of existing municipal zoning; and
- Identifying resources most susceptible to man's use.

Determining the Presence and Sensitivity of Natural Resources

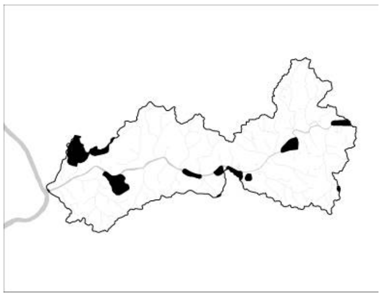
Using background mapping and input from the steering committee, public meetings and other stakeholders, the Plan identified several sensitive natural resources within the Study Area Watershed, which are relevant to potential conservation and management efforts. These resources include:

- Biological Diversity Areas;
- Wetlands;

- Flood Prone Areas;
- Slopes greater than 25 percent;
- Prime and Good Warm Water Fish Habitats; and
- Exceptional/High Quality Watersheds.

The significance and sensitivity of these natural resources were evaluated based upon a review of county, regional and state environmentally-based mapping and planning studies. After an inventory of each resource's specific location was mapped, it was important to develop an understanding of the relationship between these resources.

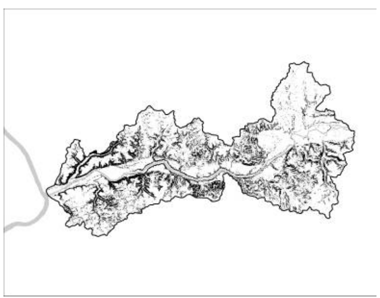
A sampling of these resources include:



Biological Diversity Areas (BDAs)

Definition: Areas (habitats) that contain significant or exceptional species, relatively large diversity of species, or entire communities or ecosystems.

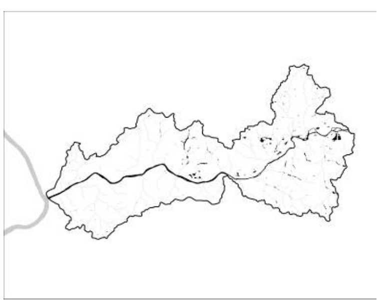
Importance: BDAs encompass the most critical habitat areas for a wide range of species in the area and represent opportunities for conservation or protection from development because they are currently relatively undisturbed by human activity.



Steep Slopes (greater than 25 percent)

Definition: Area of the landscape whose gradient is greater than 25%.

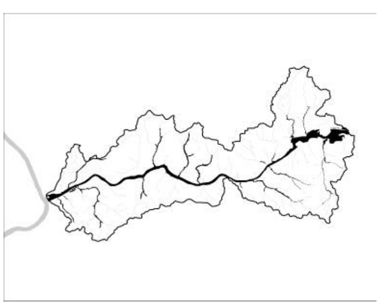
Importance: Steep Slopes, especially after human disturbances, are highly susceptible to landslides and erosion, and the ensuing loss of vegetation, aquatic habitat, water quality and property that can take place.



Wetlands

Definition: Areas of land that are frequently covered by ground or surface water that support vegetation and aquatic life.

Importance: Many species can only survive in the precise balance of a wetland ecosystem. Wetlands filter pollutants from runoff water and balance the discharge of rainwater into streams and rivers.



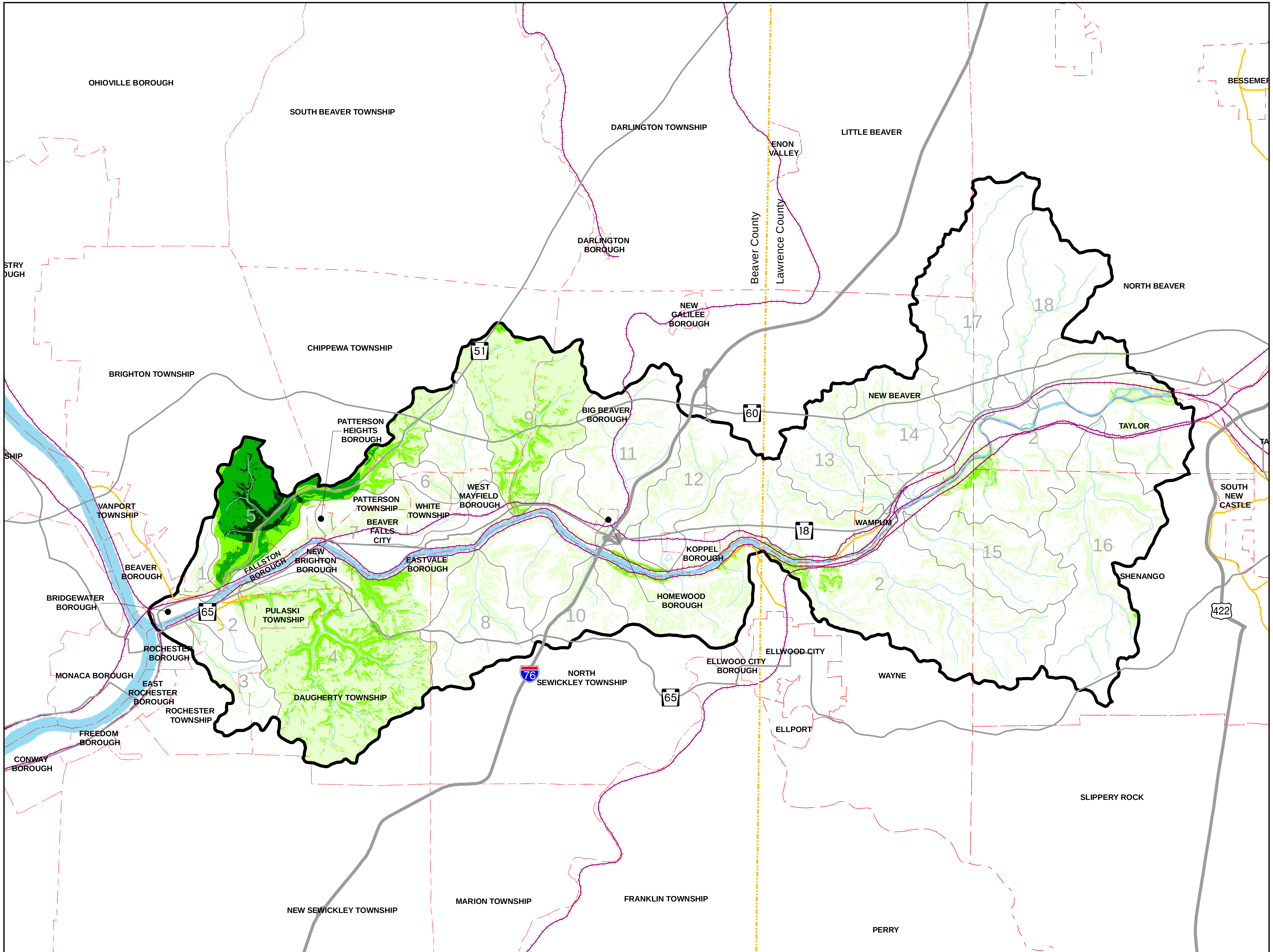
Flood Prone Areas

Definition: Areas of land that have a one-percent chance of being flooded in any given year.

Importance: Flood prone areas hold a river's natural overflow in the event of heavy rain. They allow the river to maintain its overall health by spreading temporarily. Human settlement in these areas risks being inundated.

The actual relationships of these resources within the Study Area are illustrated on the following Environmentally Sensitive Resources Composite Map.

The map presents these environmentally sensitive resources as a series of layers where each layer is given an equal value, or “weight,” of importance. When more than one resource exists in the same location, darker shading is illustrated. Consequently, the map’s lighter areas contain fewer overlapping resources while darker areas represent overlaps between a greater number of resources.



Legend

- Study Area Boundary
- Watershed Boundary
- County Boundary
- Municipal Boundaries
- Highways
- Major Roads
- Rivers/Streams/Lakes

Environmentally Sensitive Resources
(Total Number of Features)

- 0
- 1
- 2
- 3
- 4

Environmentally Sensitive Resources

1. Biological Diversity Areas
2. Wetlands
3. Flood Prone Areas
4. >25% Slopes
5. Prime Warm Water Fish Habitat
6. Good Warm Water Fish Habitat
7. Exceptional/High Quality Watersheds (Presence of Exceptional/ High Quality Streams. Includes streams with trout stock potential.)

Sub-Watersheds

1. Hamilton Run
2. Beaver River
3. McKinley Run
4. Block House Run
5. Brady Run
6. Grimms Run
7. Walnut Bottom Run
8. Bennett Run
9. Wallace Run
10. Thompson Run
11. Clarks Run
12. Stockman Run
13. Wampum Run
14. Eckles Run
15. Snake Run
16. McKee Run
17. Jenkins Run
18. Edwards Run

1,000 Acres
500 Acres
250 Acres

ENVIRONMENTALLY SENSITIVE RESOURCE COMPOSITE ANALYSIS

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: September 2006
1989.06.13r1

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NORTH

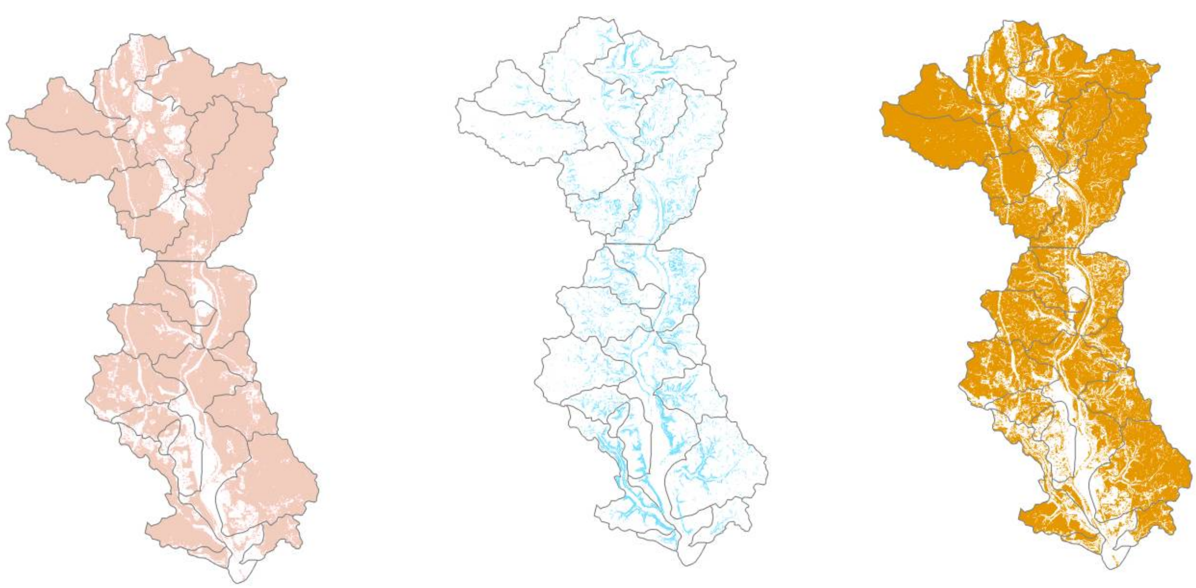
Impacts of Existing Municipal Zoning

In addition to assessing the patterns of existing resources, each community's general zoning designations were overlain with existing land coverage information to determine what the potential results of "build-out" on the Beaver River Study Area could be. In communities where zoning does not exist, a general assessment of existing intensity of development has been evaluated in context of undeveloped land in order to study such impacts of build out. Build-out refers to the point in time when a community reaches the development threshold it is willing to sustain. A Build-out Analysis serves as the tool for exploring and evaluating potential development implications and defining objectives for future development and conservation. Several elements are part of the Build-out Analysis equation. Specific to the Beaver River Conservation and Management Plan, these elements include:

Element 1: Existing Developable Land
Element 2: Environmental constraints (e.g. steep slopes, floodplains) regulated by code/ordinance
Element 3: Buildable acres

Simply, the analysis takes Element 1 (Study Area watershed) and subtracts Element 2. The total is the amount of buildable land. The acreage of buildable land is multiplied by the ordinances' permissible densities (or by the general assessment of community development intensity, if no ordinance exists). The result is the potential number of additional dwelling units that could be developed in the watershed. A graphic representation of this process as it applies to the Beaver River Study Area is illustrated in the following diagram.

Buildable Area Diagram



Existing Developable Land - Environmental Constraints = Buildable Acres

Approximately one-fourth (14,000 acres) of land in the Study Area Watershed is currently developed and contains 35,000 dwelling units. That translates to an average density of just over 2.5 dwelling units per acre of existing development. The build-out analysis examines the question "if all remaining land were developed at the density currently permitted by each of the communities' individual zoning ordinances or the continuation of the past density trends, what could the impact of population and development patterns be?"

Based upon the range of permissible development intensities applicable to the remaining undeveloped land in the Study Area Watershed, four categories of future development could occur:

Rural dwelling areas – 2-acre minimum lot size;
Suburban dwelling areas – 1-acre minimum lot size;
Low urban dwelling areas – ½-acre minimum lot size; and
High urban dwelling areas – ¼-acre minimum lot size.

The build-out analysis accounts for more than 5,000 acres of land on which environmental constraints (steep slopes, floodplains and the like) exist. In considering these factors, the analysis identified that existing zoning could permit approximately 28,000 additional households to be constructed in the Study Area. One example of working through this calculation can be seen in North Sewickley Township. Based on available general land use data and aerial photography, approximately one-third of the community is currently developed with 2,300 homes. About 5,300 acres within the community remain undeveloped. However, in considering environmental constraints, the total buildable area becomes 4,600 acres. Based upon general planning regulations and the current patterns of development, the analysis identifies that approximately an additional 2,300 homes could be constructed – identifying that the population of North Sewickley has the potential to essentially double if the community were to be built-out.

Notably, the communities which permit the greatest amount of growth based on existing zoning include the following six communities. These communities contribute nearly 70 percent of the Study Area Watershed's potential number of units.

New Beaver Township	Daugherty Township
Shenango Township	North Sewickley Township
Big Beaver Township	North Beaver Township

However, in looking at the potential scale of growth, the communities which could be most intensely developed include the following four communities.

Beaver Falls	Koppel Borough
New Brighton Borough	Patterson Heights Borough

Table 2-1 presents the build-out analysis' results per community and for the overall Study Area.[1]

[1] Chapter 4 has more information about land use regulations by municipality.

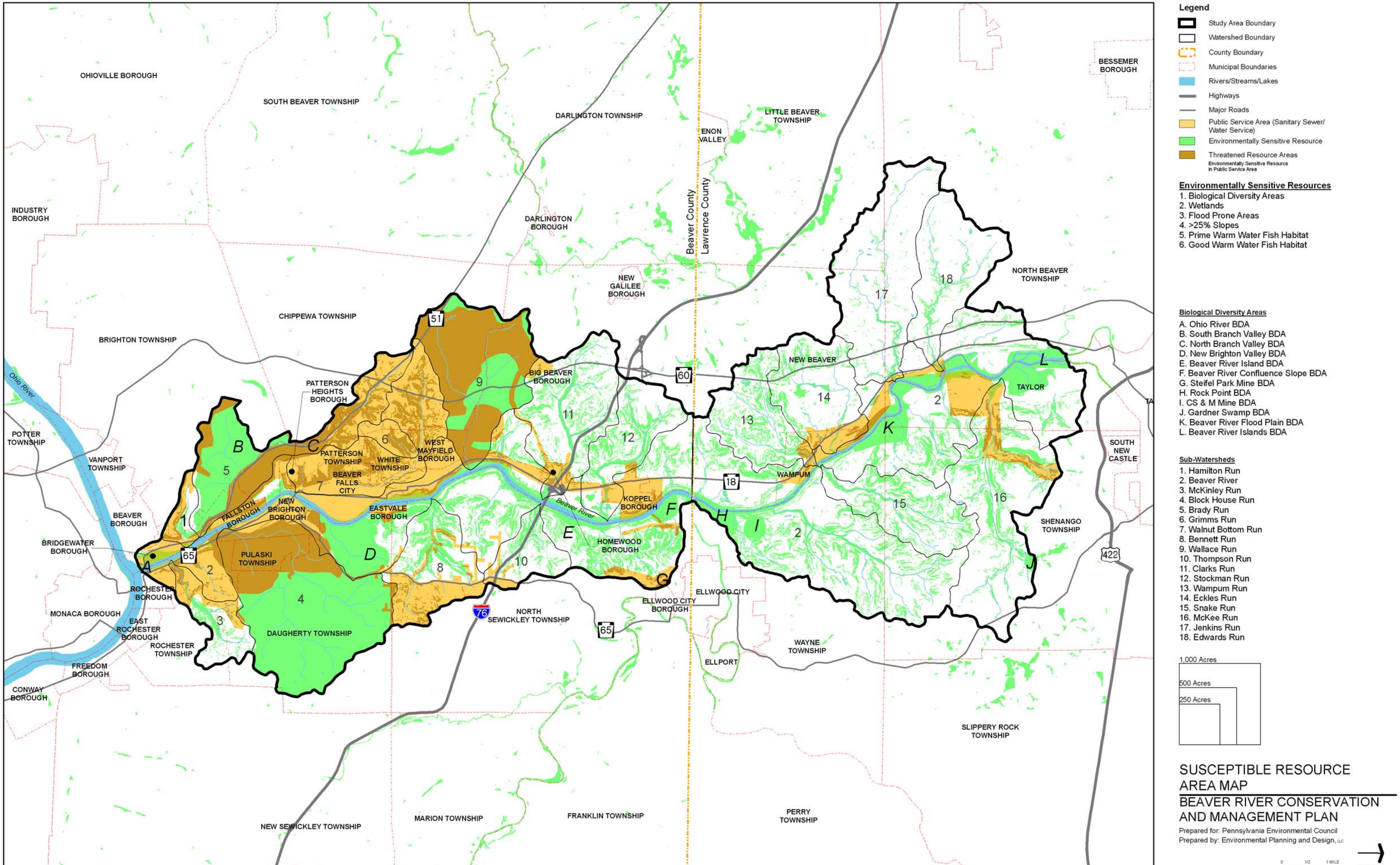
Table 2-1: Build-out Analysis based on Generalized Zoning

	Communities	EXISTING		BUILDABLE AREA			PROJECTED	
		Developed Watershed Acres	Existing Dwelling Units	Total Undeveloped Watershed Acres	Undeveloped Watershed Acres with Environmental Constraint	Total Buildable Acres	Total Projected Dwelling Units for Buildable Acres	Total Dwelling Units at Buildout of Watershed Acres
1	BEAVER BOROUGH	19	2,297	1	1	0	0	2,297
2	CITY OF BEAVER FALLS	1,183	4,380	275	93	182	728	5,108
3	BIG BEAVER TOWNSHIP	1,162	905	6,099	780	5,319	2,660	3,565
4	BRIDGEWATER BOROUGH	270	335	140	76	64	132	467
5	BRIGHTON TOWNSHIP	185	2,875	1,223	180	1,043	1,043	3,918
6	CHIPPEWA TOWNSHIP	1,265	2,933	2,034	205	1,829	1,829	4,762
7	DAUGHERTY	769	1,317	4,268	443	3,825	3,825	5,142
8	EASTVALE BOROUGH	65	133	13	5	8	15	148
9	FALLSTON BOROUGH	175	136	142	94	48	48	184
10	HOMEWOOD BOROUGH	68	62	42	16	26	13	75
11	KOPPEL BOROUGH	292	409	49	17	32	130	539
12	LITTLE BEAVER TOWNSHIP	1	502	612	4	608	304	806
13	NEW BEAVER TOWNSHIP	791	696	5,265	314	4,951	4,951	5,647
14	NEW BRIGHTON BOROUGH	673	2,999	43	28	15	60	3,059
15	NORTH BEAVER TOWNSHIP	757	1,580	4,994	461	4,533	2,266	3,846
16	NORTH SEWICKLEY TOWNSHIP	1,042	2,326	5,351	713	4,638	2,774	5,100
17	PATTERSON HEIGHTS BOROUGH	92	268	67	41	26	103	371
18	PATTERSON	558	1,332	531	197	334	668	2,000
19	PULASKI TOWNSHIP	304	764	223	59	164	196	960
20	ROCHESTER BOROUGH	168	1,900	1	1	0	0	1,900
21	ROCHESTER	754	1,268	536	123	413	254	1,522
22	SHENANGO TOWNSHIP	770	2,996	6,458	454	6,004	3,002	5,998
23	TAYLOR TOWNSHIP	1,044	506	1,637	473	1,164	1,433	1,939
24	WAMPUM BOROUGH	605	310	141	65	76	84	394
25	WAYNE TOWNSHIP	441	946	3,800	510	3,290	1,645	2,591
26	WEST MAYFIELD BOROUGH	327	499	174	27	147	209	708
27	WHITE TOWNSHIP	299	667	165	71	94	254	921
	Totals	14,079	35,341	44,283	5,451	38,832	28,625	63,966

Susceptible Resource Area Map

Within the Study Area, lands examined as part of the build-out analysis were further evaluated to determine where there are overlaps between known sensitive resources and existing access to infrastructure systems. Without strategic planning, this relationship could create the potential for resources to be at risk, or susceptible to unwanted impacts. The Susceptible Resource Areas Map illustrates instances where these overlaps occur.

Extensive areas of susceptibility include the portions of Brady's Run watershed, a majority of the Wallace Run watershed, the western section of the Block House Run watershed and various sections of the Walnut Bottom Run watershed. The manner in which the potential impacts of human use, development and/or infrastructure expansion could be managed are identified in Chapter 3.



SUSCEPTIBLE RESOURCE AREA MAP

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Modeling the Impacts of Flood Events

A unique aspect of the Beaver River Conservation and Management Plan is the use of computer modeling to identify the extent to which flooding may impact or may be impacted by future growth. The two primary variables within the Study Area which potentially influence the effects of future flood events are:

1) The amount and intensity of development; and
2) The nature and extent of policies regulating this development.

Outlined below are the model's assumptions, base conditions, and findings.

Assumptions and Base Conditions

As part of the analysis process, general stormwater impacts were assessed by incorporating available existing regional land coverage data within a flood model program (HEC-RAS®) developed through the US Army Corps of Engineers. Historical flood data were mapped to establish the scale and extent of past flooding events. This modeling incorporates the land coverage data with the flood data (including severe weather/hurricane events of 2004 [i.e. Hurricane Ivan]) to establish the basic relationship between existing development and stormwater run-off in the Study Area's various watersheds. For portions of the Study Area where background data were unavailable, a series of hydrological factors replicating patterns of development intensity and known storm event water levels were integrated into the flood model.

Future land use information was also examined and "tested" to evaluate if and how potential development at build-out[2] could contribute to future flooding levels. Build-out information was not available for communities upstream of the Study Area or within the Connoquenessing Creek watershed (which flows from the east into the Beaver River). Consequently, impacts of any additional development beyond the Study Area have not yet been evaluated as part of this modeling.

As part of the background analysis efforts, known conditions were identified at seven sample locations along the Beaver River and along selected tributaries. Each location possesses a slightly different character in terms of the water's known width, depth, and the type and intensity of existing surrounding development. For each location, four critical components were examined:

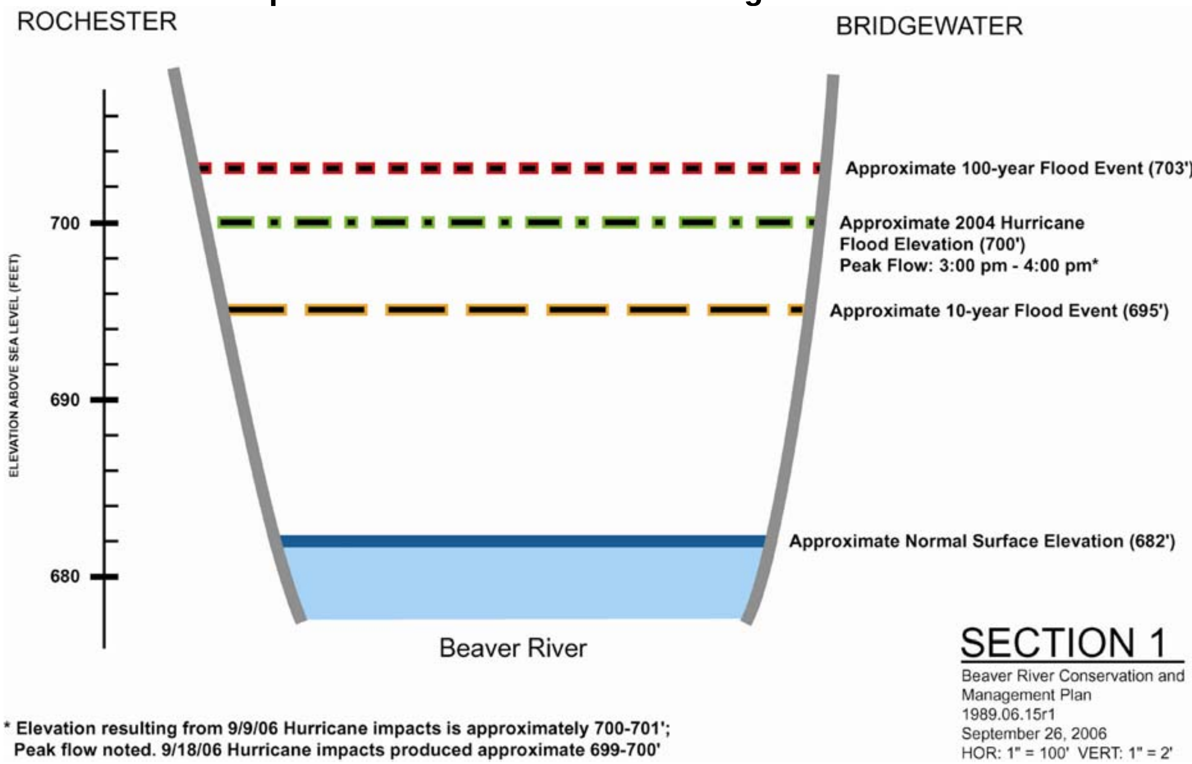
a. Normal (water) surface elevation;
b. Approximate 10-year flood event elevation;
c. Approximate 100-year flood event elevation;
d. Approximate 2004 hurricane flood elevation.

To compare each location's similarities and differences, a "cross section," or imaginary "slice," of the river and selected tributaries was prepared. These sections are found on the following pages. Each critical component is represented by a line of varying pattern and color. Normal surface elevation represents the typical daily water elevation, or existing surface water conditions. 10-year and 100-year flood events typify occasions when an area will be inundated by a flood event that has a 10 percent chance or one percent chance, respectively, of being equaled or exceeded in any given year. The line illustrating the 2004 hurricane flood elevation represents the average elevation the water reached during the series of hurricane-related flood events impacting southwestern Pennsylvania in the Fall of that year. Elevations provided are based upon existing development

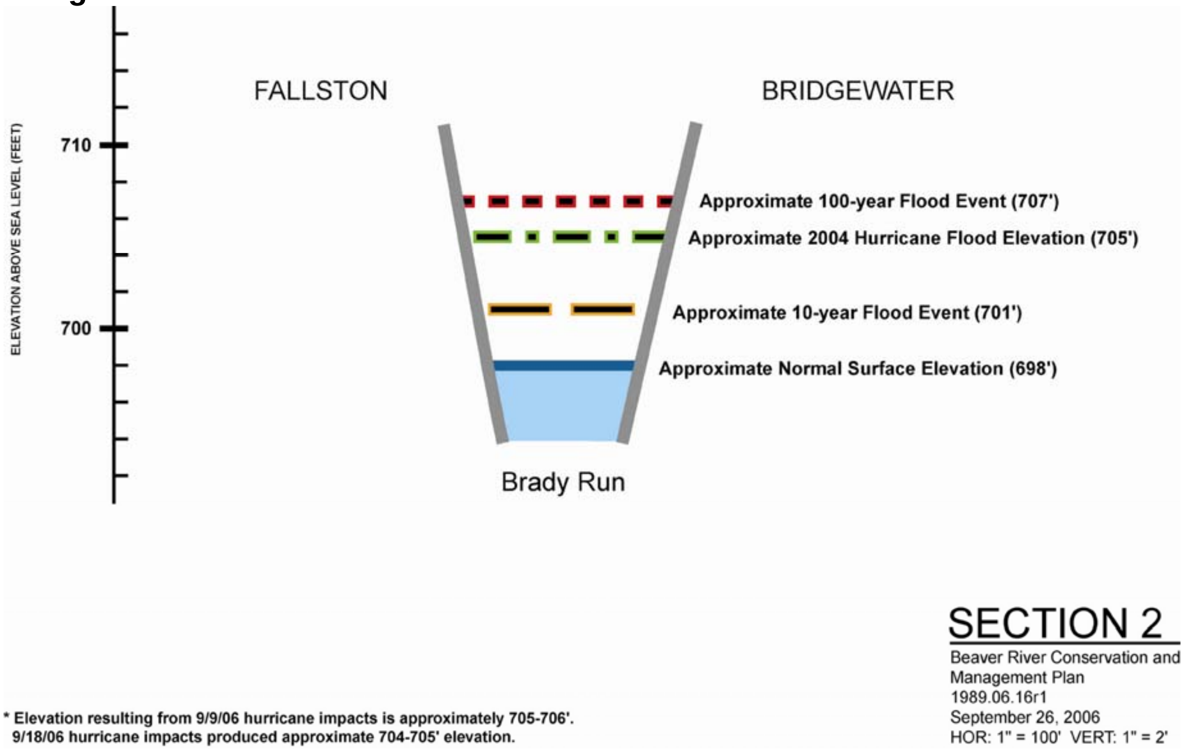
[2] all development which is permitted to occur by the Study Area communities' Ordinance provisions

patterns and intensities. Notably, in several studied locations, impacts of the 2004 hurricane-related floods produced effects which fall between 10-year and 100-year flood stages.

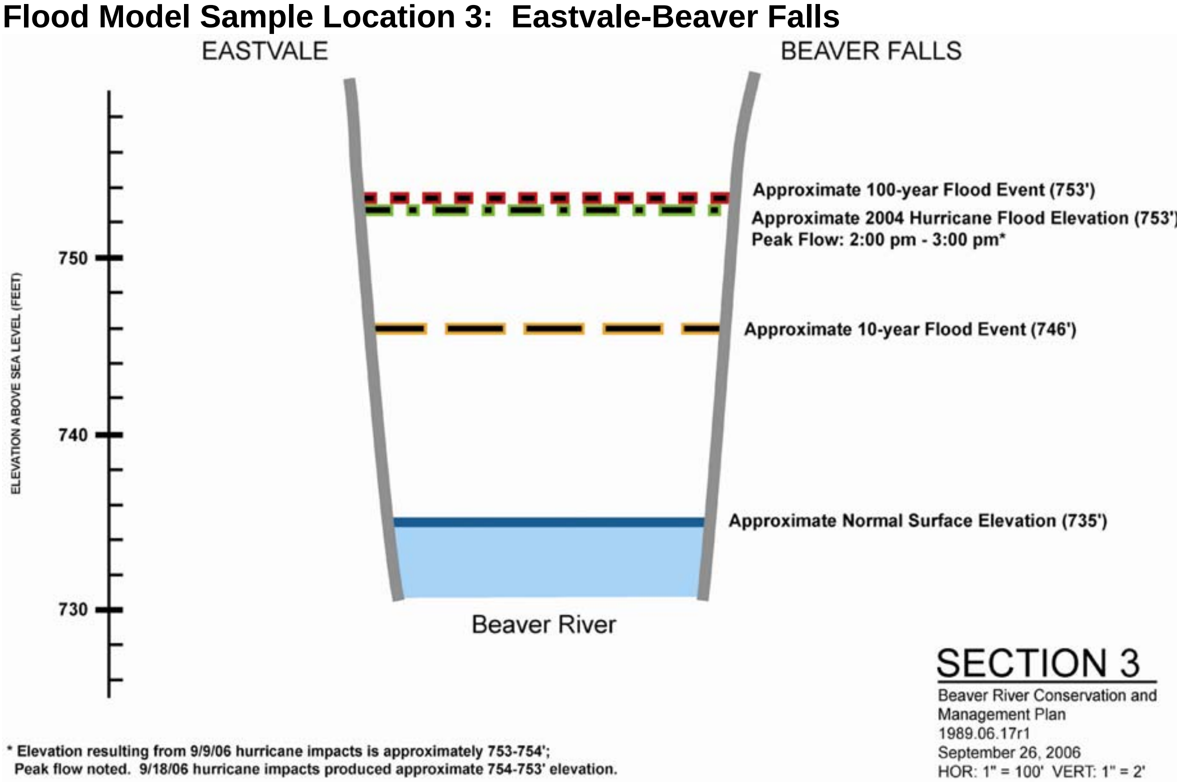
Flood Model Sample Location 1: Rochester-Bridgewater



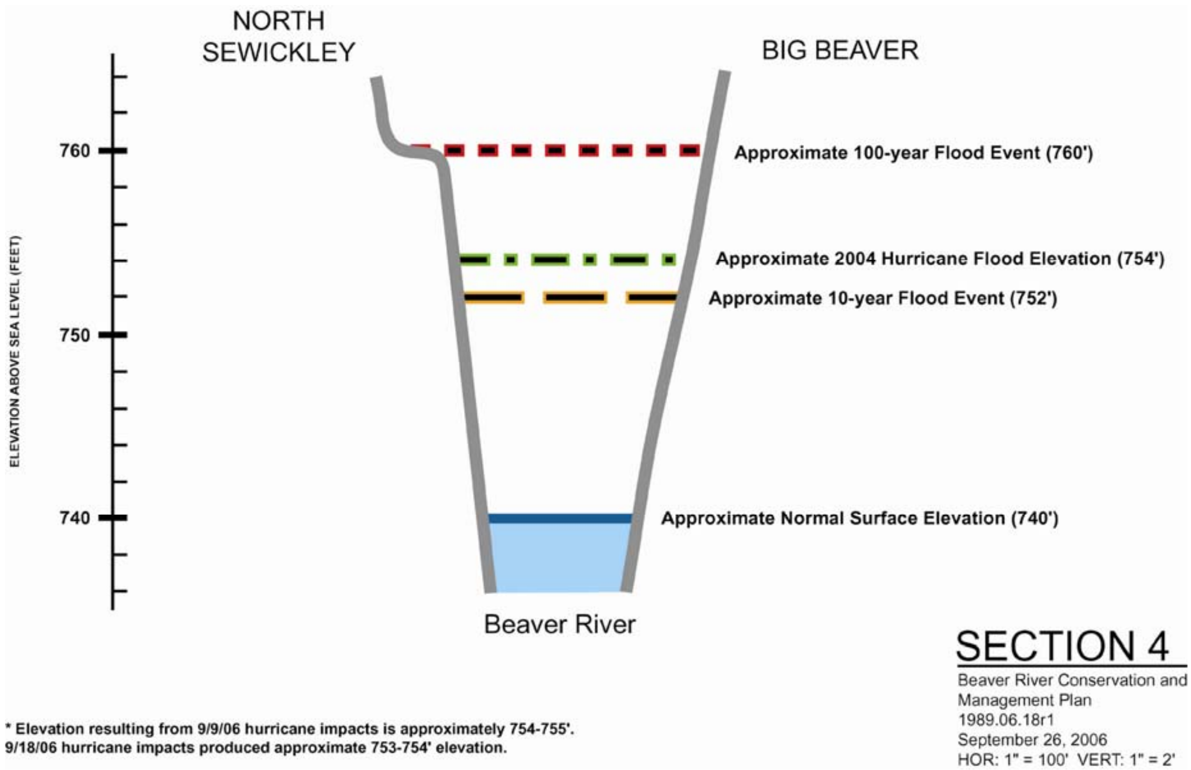
Flood Model Sample Location 2: Fallston-Bridgewater



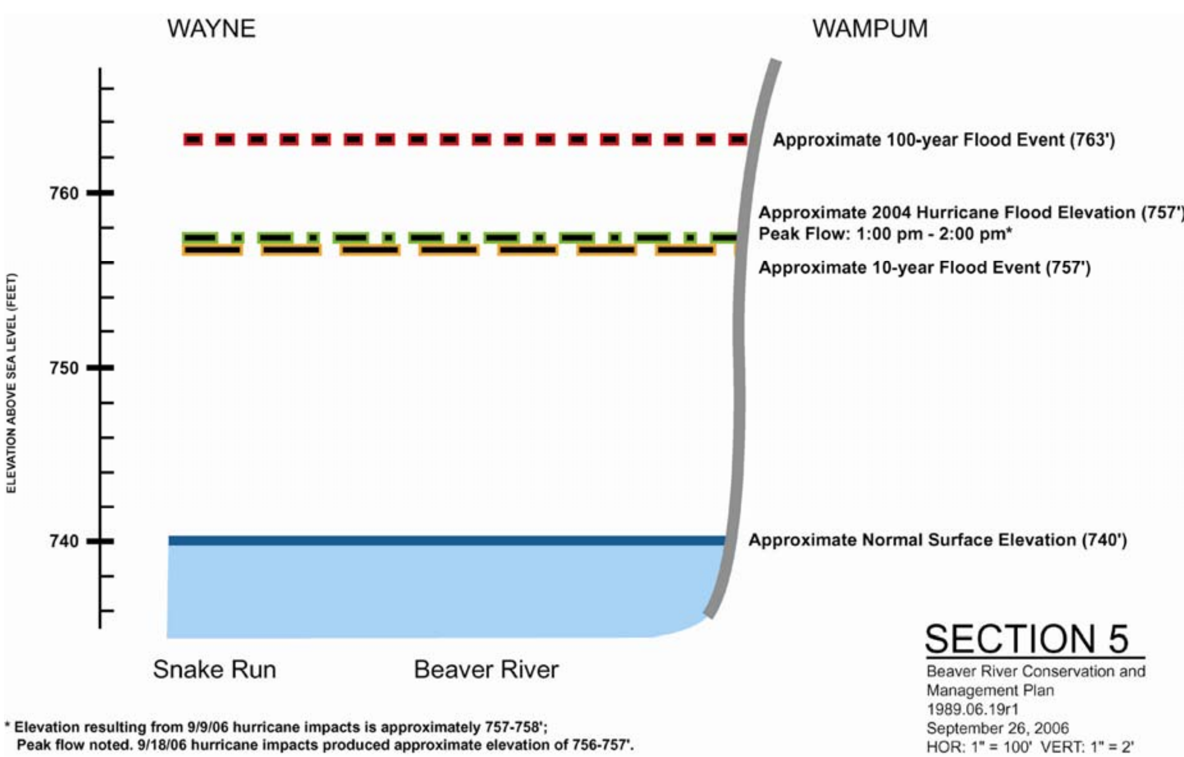
Flood Model Sample Location 3: Eastvale-Beaver Falls



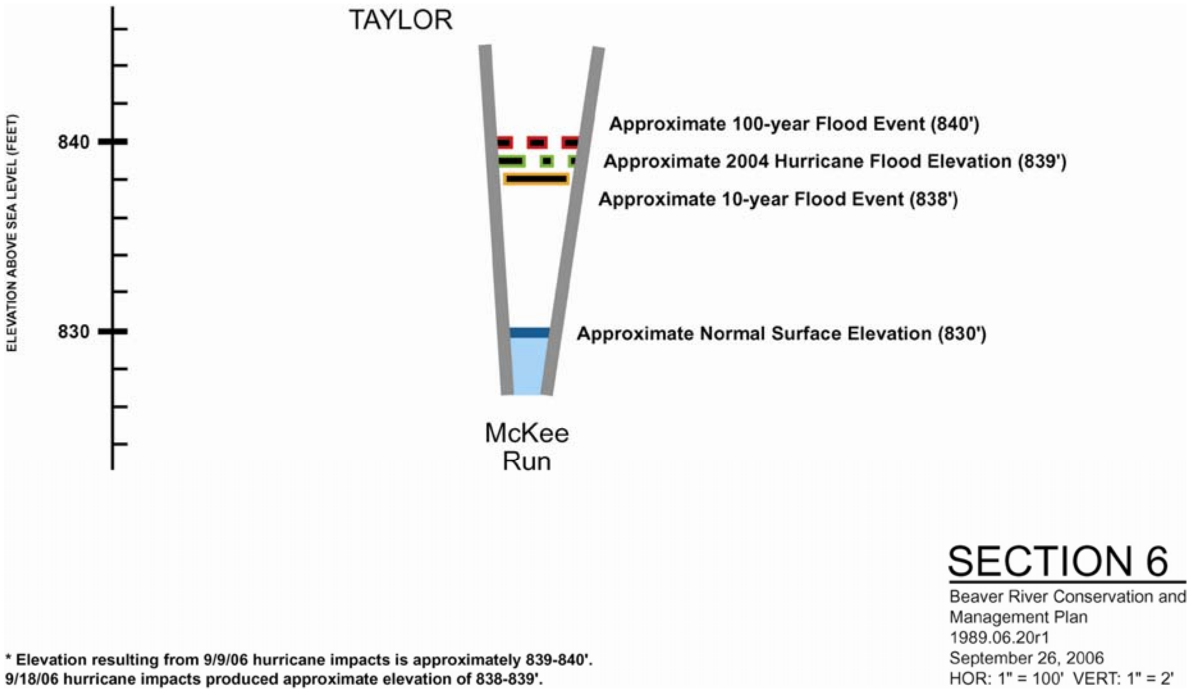
Flood Model Sample Location 4: North Sewickley-Big Beaver



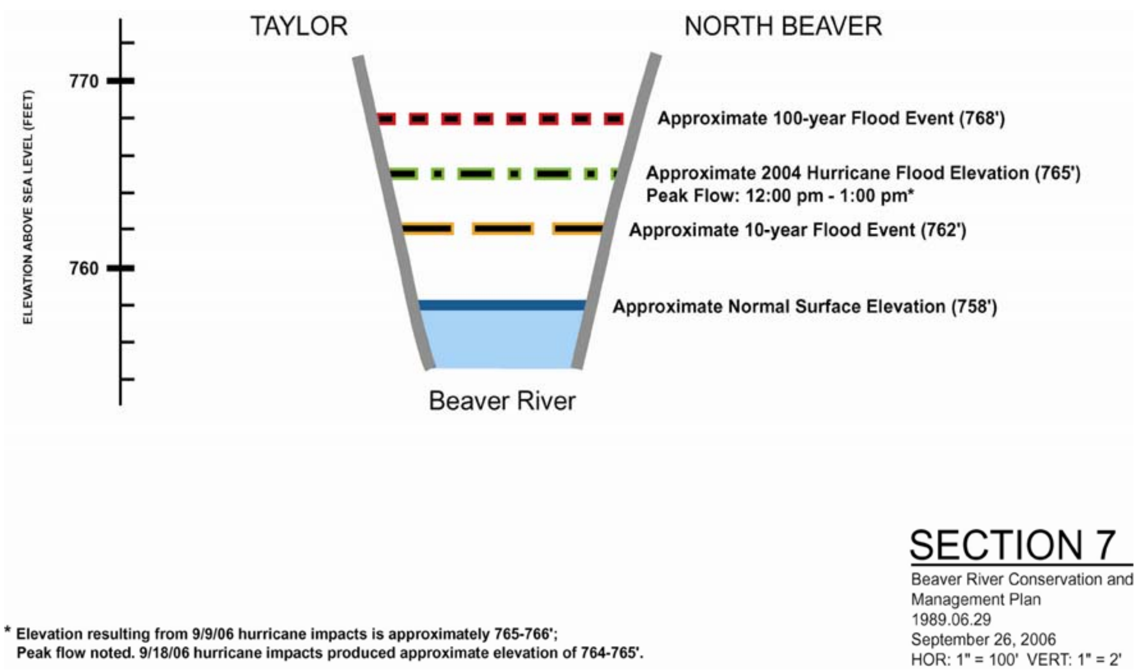
Flood Model Sample Location 5: Wayne-Wampum



Flood Model Sample Location 6: Taylor



Flood Model Sample Location 7: Taylor-North Beaver



Findings

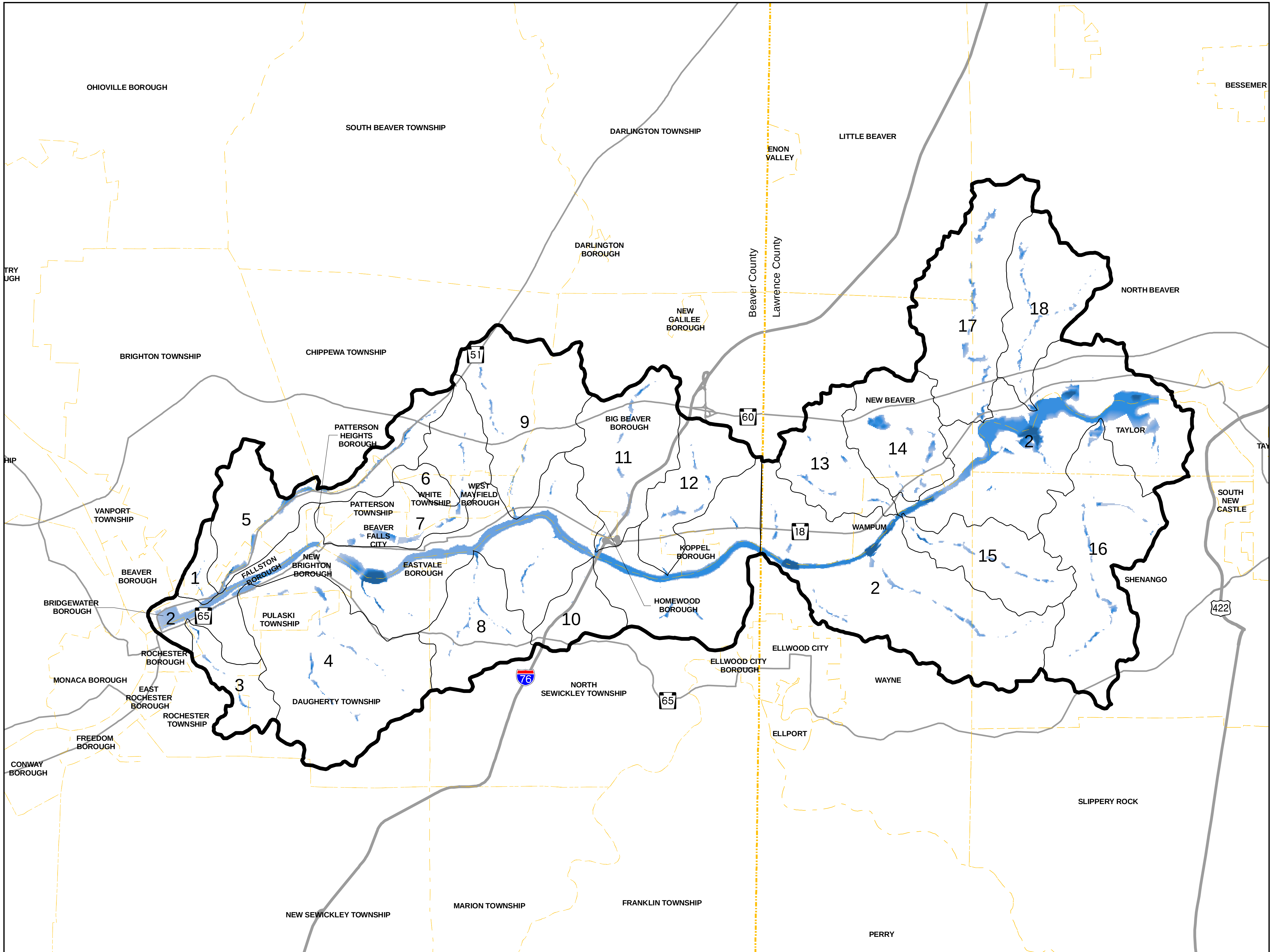
If future growth within the Beaver River Study Area were to reach build-out based upon the communities' current permitted development intensities and existing policies (or absence thereof) guiding it, the model anticipates flooding levels will increase from those experienced today. Looking at the overall Study Area, it is likely that build-out could raise typical flooding events by a minimum of 1 to 2 feet. In watersheds currently approaching build-out, minimal change to flooding patterns is anticipated. In portions of the Study Area where the amount and intensity of development may be more substantial, it is most likely that greater rises in future flooding could also result.

During severe weather events, more intense impacts will also likely occur in streams which are generally narrow, shallow and/or are comprised of steeper slopes. Also, increases in flooding are more likely in bodies of water which are in watersheds north of the Pennsylvania Turnpike.

To further illustrate analysis results, a series of diagrams were generated. The Base Flood Condition Map depicts typical seasonal flooding levels at build-out. Where darker shading occurs in a water body, flooding is anticipated to be more extensive. Factors such as future development intensity and topography along the stream's/river's boundary contribute to this result.

The Severe Event Flooding Conditions Map illustrates the change in potential flooding during a severe weather event such as 2004's Hurricane Ivan. Coupling additional water flow with additional development results in broader areas of the watershed being inundated with higher water. This is evidenced through more extensive portions of darker shading (high and moderate inundation) than as seen on Base Flood Map.

Both scenarios assume existing development regulations (e.g. impervious coverage, permitted densities, etc.) remain as currently permitted. As identified in later portions of this Plan, opportunities to reduce the extent and/or impacts of potential flooding may be seen through the introduction and adoption of alternative resource management techniques and development provisions.



Legend

- Study Area Boundary
- Watershed Boundary
- County Boundary
- Municipal Boundaries
- Highways
- Major Roads

FLOOD INUNDATION (Feet Above Existing Pool)

High

Low

1. Hamilton Run
2. Beaver River
3. McKinley Run
4. Block House Run
5. Brady Run
6. Grimms Run
7. Walnut Bottom Run
8. Bennett Run
9. Wallace Run
10. Thompson Run
11. Clarks Run
12. Stockman Run
13. Wampum Run
14. Eckles Run
15. Snake Run
16. McKee Run
17. Jenkins Run
18. Edwards Run

1,000 Acres

500 Acres

250 Acres

BASE

FLOODING CONDITIONS

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvannia Environmental Council

Prepared by: Environmental Planning and Design, LLC

Date: February 2008

1989.08.01

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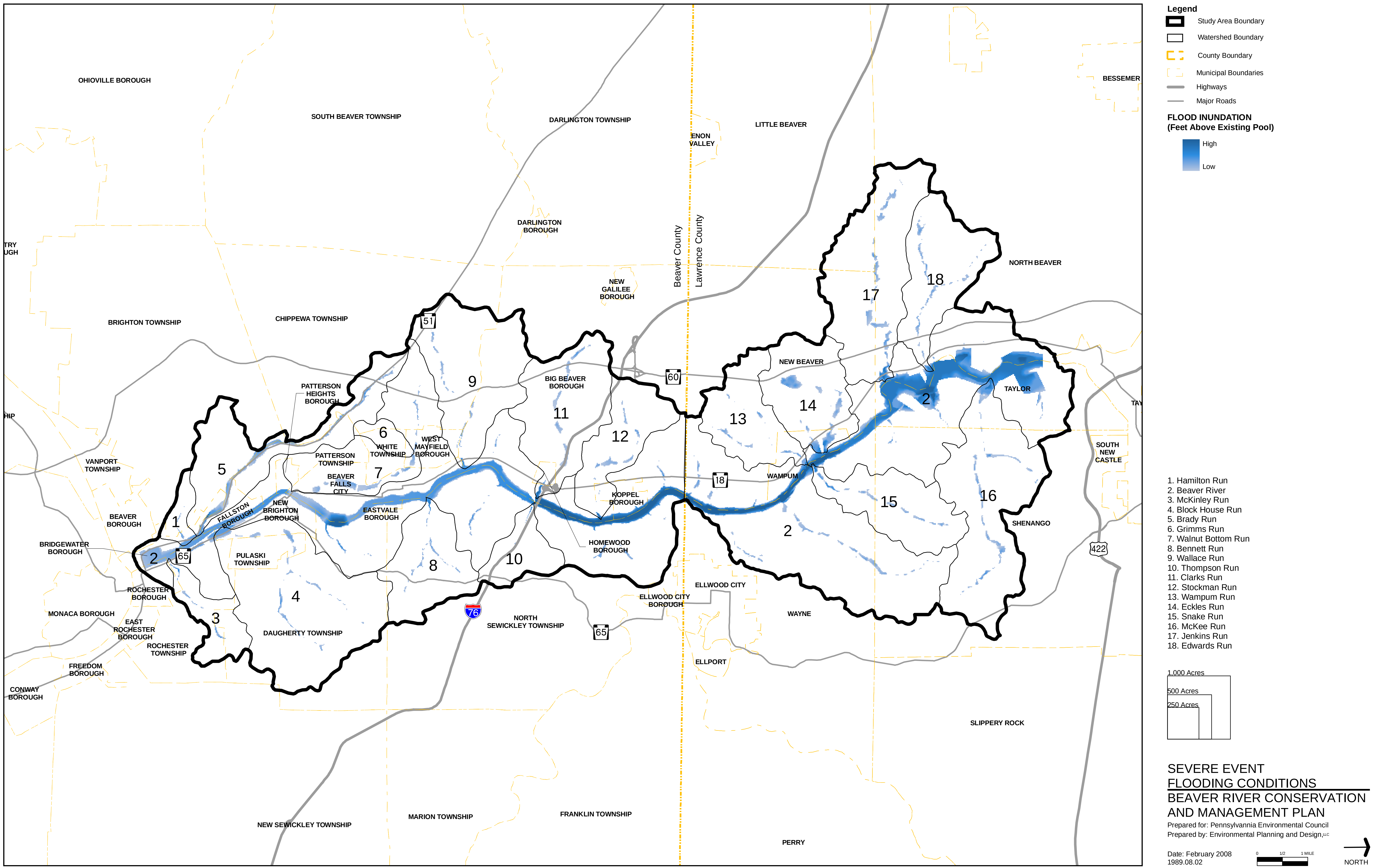
NORTH



SEVERE EVENT
FLOODING CONDITIONS

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvannia Environmental Council
Prepared by: Environmental Planning and Design,LLC

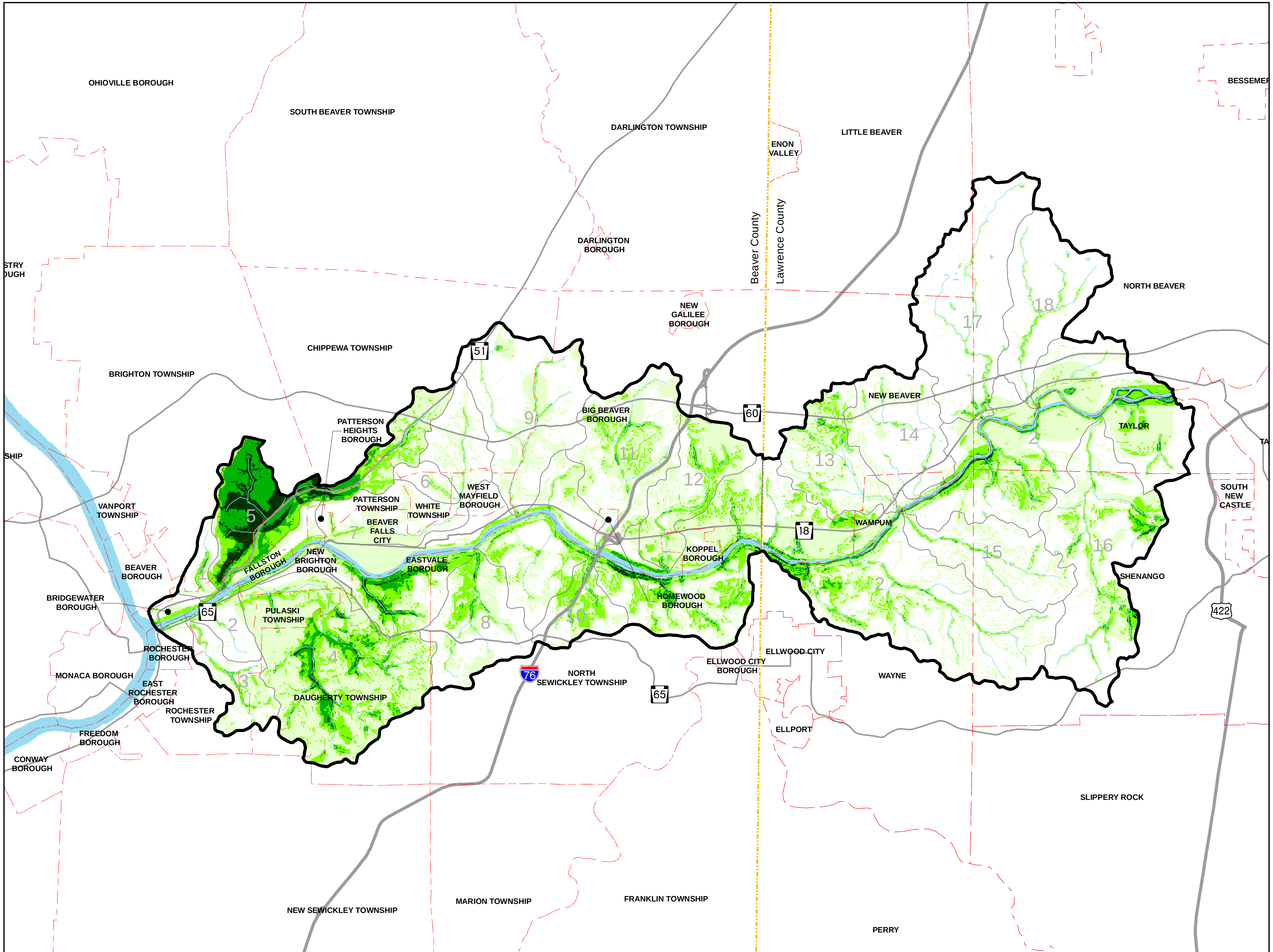
Date: February 2008
1989.08.02

Conservation Strategy Analysis

The compilation of all project background work comes together in the Conservation Strategy Analysis. The Conservation Strategy Analysis Map identifies the location and intensity of resources which residents and leaders of Beaver and Lawrence Counties have identified to be most significant for future recognition as related to conservation and/or management. A matrix on the right hand side of the map includes three general resource categories: Environmentally Sensitive Lands, Habitat and Recreation. Each category contains a series of specific resources. For example, trails are categorized under Recreation and woodlands are categorized under Habitat.

For each resource recognized on the matrix, its significance and sensitivity were determined by the Pennsylvania Environmental Council with input from the Advisory Committee. These were then “weighted” based on perceived importance, resulting in a “score” for each resource. The resource was assigned a corresponding shade of green. A resource with greater sensitivity to potential impacts of man-made activities and/or development is assigned a higher score. The sum of the resources’ scores, as many overlap, affects the map’s patterns of shading. As the sum increases, the shading darkens. It is important to note that an area’s sensitivity is measured by the sum and not the quantity of resources. For example, one resource with a weighted sum of 4 is more sensitive than three resources each with a weight of 1, summing to a score of 3.

Based on this analysis, areas with the greatest identified significance and sensitivity include portions of eastern Brighton Township, the northern Bridgewater Borough and a small section of southeastern Chippewa Township. These municipalities are generally within the Brady’s Run tributary watershed.



Legend

- Study Area Boundary
- Watershed Boundary
- County Boundary
- Municipal Boundaries
- Highways
- Major Roads
- Rivers/Streams/Lakes

TOTAL SCORE

- 1 - 5
- 6 - 10
- 11 - 15
- 16 - 28

Conservation Matrix

		Resource	Weighted Value
Environmentally Sensitive Lands	a	Exceptional/High Quality Watersheds (Presence of Exceptional/High Quality Streams. Includes streams with trout stock potential.)	5
	b	Wetlands	5
	c	Steep Slopes (>25%)	4
	d	Stormwater Filtration	4
	e	Floodplains	2
	f	Groundwater Recharge	1
Habitat	a	BDAs	5
	b	Habitable Woodlands	5
	c	Warmwater Gamefish Habitats	3
	d	Important Bird Areas	3
Recreation	a	Trails	3
	b	Non-motorized Boating/Fishing Access (combines flatwater, whitewater and sail boating)	3
	c	Primitive Camping	2
	d	Hunting	2
	e	Motorized Boating/Fishing Access	1
	f	Non-Primitive Camping	1
	g	ATV areas	0

Sub-Watersheds

1. Hamilton Run
2. Beaver River
3. McKinley Run
4. Block House Run
5. Brady Run
6. Grimms Run
7. Walnut Bottom Run
8. Bennett Run
9. Wallace Run
10. Thompson Run
11. Clarks Run
12. Stockman Run
13. Wampum Run
14. Eckles Run
15. Snake Run
16. McKee Run
17. Jenkins Run
18. Edwards Run

1,000 Acres
500 Acres
250 Acres

CONSERVATION STRATEGY ANALYSIS

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: September 2006
1989.06.25r1

0 1/2 1 MILE

NORTH

Observations

1.) Development and Infrastructure
The Beaver River Study Area Watershed is characterized by a combination of developed and undeveloped land. Sewer and water infrastructure services both of these landscapes. Consequently, the potential exists for some undeveloped or very low intensely developed areas that possess environmentally sensitive resources and the availability of infrastructure to incur pressures of development. Through assessing the relationships between existing development, existing zoning ordinance provisions and potential build-out patterns, it is possible that if and/or when all land within the communities of the Beaver River Study Area Watershed develop, the Corridor's population and number of dwelling units could nearly double. A key consideration in this matter is in the understanding that the communities' current ordinance development-related provisions permit this type of growth. Considerations of growth also should be evaluated in context of the communities' and region's overall planning objectives.

Another consideration of existing development and infrastructure patterns is the implication of flooding, both normal and severe conditions. Flooding is typically a response to a combination of factors including development patterns and impervious conditions, the presence and adequacy of stormwater management controls, and climatic changes. Impervious conditions are those in which water cannot freely flow into the ground. Through the assessment of these characteristics, as related to the Beaver River Study Area Watershed, there are notable times in history in which flooding has impacted both lands along the Beaver River's banks as well as in other portions of tributary watersheds. The potential patterns of development in the Study Area provide the opportunity to alter these types of patterns in the future. Depending upon the stormwater management mechanisms and policies promoted and implemented within the Corridor in the future, the impacts from instances of flooding could either improve or become more significantly impacted than the patterns of today.

2.) Management and Outreach
Currently, most municipalities within the Beaver River Study Area Watershed are charged with evaluating and approving/denying development and resource impacts within the confines of their individual borders. There is limited, if any, coordination between neighboring local government entities as related to watershed management. Some regions in Pennsylvania and in other states utilize a watershed organization to fulfill such an advisory and/or regulatory role. No such organization currently exists within the Beaver River Study Area Watershed or in the greater geographic region.

Recommendations and Policies

From the analyses and observations, recommendations and policies for future conservation and management strategies can be formulated. Each of the following recommendations for implementation has been evaluated related to current projects and policies. Each recommendation has been assessed based on its potential to enhance the physical, ecological and/or economic well-being of the Beaver River Study Area Watershed. The Plan's recommendations, outlined in the following list, are generally related to one of three categories:

(1) Management
(2) Development and Infrastructure
(3) Outreach, Education and Public Awareness

Local, regional, and county-wide entities currently serve the communities of the Study Area in a variety of ways. It is recommended that the key stakeholders initiating implementation efforts associated with this Rivers Conservation and Management Plan be identified from groups leading land use planning efforts in both Beaver and Lawrence Counties and to form partnerships to leverage resources in implementing effective policies at the local or county level. These groups should work cooperatively with local and regional leaders and organizations to identify, draft, present and implement the Plan's recommended conservation and management strategies, policies, and projects.

Consequently, to promote consistency between policies and implementation, it is recommended that Beaver and Lawrence Counties take a lead role in establishing and promoting consistent standards for resource management. Specifically, the Counties should collaborate to pursue the following as listed in order of recommended priority. Priorities have been identified based upon project analysis findings and feedback gained throughout the preparation of this plan.

1.) *Identify or form a conservation and management advisory body to implement the recommendations of the Conservation and Management Plan*

This advisory body can:

- Identify environmental problems and recommend ordinances, plans, and programs to the appropriate municipal agencies for the promotion and conservation of natural resources and for the protection and improvement of the quality of the environment within its municipal boundaries;
- Promote community environmental programming;
- Keep an index of all open space, publicly and privately owned, including flood-prone areas, swamps, and other unique natural areas, for the purpose of obtaining information on the proper use of such areas;
- Make recommendations for the possible use of open land areas;
- Advise the appropriate government agencies, including but not limited to planning commissions and elected governing bodies on the commitment of suitable property, both real and personal, for conservation;
- Assist communities in pursuing grant money for improvement projects, including those initiatives offered through the Department of Community and Economic Development (www.newPA.com).

One example of such an advisory body is an Environmental Advisory Council (EAC), authorized by Act 148 of 1973. An EAC can assist local governments in identifying natural resource opportunities and challenges and recommending plans and programs to protect and improve the quality of the community's environment. Contact the EAC Network for more information (http://www.greentreks.org/eacnetwork/index.asp).

2.) *Create a River Corridor Management Ordinance*

As part of their Act 167 Plans, County representatives should work with local municipalities to develop and adopt a River Corridor Management ordinance that addresses development, implementation, and enforcement of programs to manage stormwater from new development and re-development. In managing stormwater, watershed communities can minimize/reduce potential flooding problems, erosion, and loss of or damage to natural resources. Both counties have begun to look at stormwater management issues in the Act 167 Plans that are under development.

3.) *Adopt Impervious Surface provisions as part of the River Corridor Management Ordinance to minimize negative impacts of stormwater run-off*

Impervious surfaces are constructed areas or soils compacted by urban development which seal surfaces, repel water and prevent precipitation and melt water from infiltrating soils. Provisions which control the intensity and/or percentage of impervious surface as part of development can be utilized to decrease large volumes of runoff generated by storms as well as non-point source water pollution problems attributed to impervious surfaces. Impervious surface provisions can be adopted on either a local- or watershed-wide basis. As part of these efforts, one example that could be referenced is the City of Austin's Barton Creek Watershed Ordinance.
(www.epa.gov/safewater/sourcewater/pubs/techguide_ord_tx_austin_zoningnewdiv4.pdf).

4.) *Advocate for the creation and adoption of a Steep Slope Ordinance*

Steep slopes are defined as areas that exceed a certain percentage slope and are often associated with certain types of soils. In western Pennsylvania, steep slopes are often those that are greater than 25 or 40 percent gradient. Steep slopes are valuable resources and sensitive landforms where a diversity of organisms can thrive. Excavations or building construction can promote instability by loading the slope, removing vital support, and increasing porosity.

Some communities within the Beaver River Study Area Watershed currently limit the amount of disturbance that is permitted to occur on steep slopes. It is recommended based on the characteristics of the Beaver River Study Area Watershed, that ordinance provisions be adopted to limit the amount of disturbance that should occur on slopes greater than or equal to 25 percent and eliminate development on slopes greater than or equal to 40 percent. While the designation of different slope categories exist, one example of the range of provisions that can be addressed in a Steep Slope Ordinance include East Nantmeal, Chester County, PA. (http://dvrpc.org/planning/community/ProtectionTools/ordinances)

5.) *Evaluate the feasibility of other watershed-wide ordinance updates which coordinate development and infrastructure opportunities*

Based upon the patterns and locations of existing environmentally sensitive areas in the Beaver River Study Area Watershed, Beaver and Lawrence Counties should evaluate the feasibility of potential ordinance language and/or mapping updates which encourage the coordination of potential development and infrastructure. This relationship can promote that infrastructure investments are optimized while undesirable disturbances to existing sensitive natural resources are minimized. Based upon identified and a general evaluation of existing and known planned development opportunities, consideration for such updates should occur in many communities throughout the Corridor, specifically:

1. Brighton Township
2. Chippewa Township
3. Daugherty Township

4. New Brighton Borough
5. Patterson Township
6. Pulaski Township
7. Rochester Township

6.) Promote the development that respects the landscape's character and capacity

Counties and communities should encourage development patterns in urban, suburban, and rural areas of the Beaver River watershed that are compatible with existing character and infrastructure services while minimizing negative impacts to sensitive natural resources. Updates to land use and/or development regulations should be considered at local and regional levels to ensure implementation of coordinated best management practices at both the local and regional scales. On-going review and analysis of potential development build-out impacts, as shaped by zoning and subdivision regulations, should be completed through coordination of the County Planning Department and the newly formed river advisory body to promote coordinated conservation and development efforts throughout the region.

7.) Establish consistent review procedures to identify potential conservation needs

Based on the evaluation of potential build-out and analysis of resources potentially susceptible to disturbance, approximately 75 percent of the Beaver River Study Area Watershed could be developed. As proposals for development come about, county and regional stakeholders should work with local municipalities and the development community throughout the Corridor to establish a formal, consistent review process to identify if, when and/or how existing natural resources could be conserved. Uniform criteria for identifying such lands should also be prepared, inclusive of sensitive natural resources identified as part of this Rivers Conservation and Management Plan. Designation of conserved lands could be in the form of private and/or public open space for natural heritage or historic sites. As part of a larger regional and state-wide initiative, Illinois currently manages a series of programs including the Natural Heritage Database, the Nature Preserves System and the Natural Heritage Landmarks program which evaluate and classify lands suitable for conservation based upon a series of such criteria.

8.) Coordinate conservation with other County open space and greenway efforts

The Rivers Conservation and Management Plan presents a series of natural resource data which should be utilized as future Beaver and Lawrence County recreation and conservation greenway planning efforts are pursued.

The Beaver County Greenways Plan identifies twelve potential greenways. Of these, the Brady's Run watershed area and the southern portion of the Beaver River Study Area Watershed possess areas of sensitive natural resources that are commonly identified in both plans.

The Lawrence County Greenway Plan identifies eight potential conservation greenways. Included among these eight are the Beaver/Mahoning River Greenway, which includes corridors along Jenkins Run and Edwards Run, and the McKees Run Greenway. These areas have been identified in the Beaver River Conservation and Management Plan as having sensitive natural resources, particularly areas of dense forests.

Based upon the resources' identified significance and sensitivity to recreation and conservation, the Counties and identified stakeholders should work to coordinate how projects, policies, and funding opportunities can be optimized and implemented succinctly. Already, the Counties have a joint Greenways coordinator to help facilitate this process.

9.) *Initiate discussions with local land trusts and conservancies to conserve lands located in the River Corridor*

The leaders and residents of Beaver and Lawrence Counties should initiate discussions with local land trusts and conservancies to identify how existing scenic, recreational, educational and environmental landscapes within the Beaver River Study Area Watershed could offer opportunities for land conservation and promoting healthier watersheds.

Building on these priorities, the following Action Plan provides the full range of recommendations for Beaver and Lawrence Counties, local municipalities and regional organizations to pursue in enhancing conservation and management efforts within the Study Area. The recommendations were developed based on input from the Advisory Committee, public meetings, surveys of stakeholders, and reviewed by the Pennsylvania Environmental Council and their consultant, Environmental Planning and Design.

CHAPTER 3: ACTION PLAN

The Action Plan is a summary of the implementation steps, or actions, necessary to complete the Plan's major recommendations. Actions vary from recommendation to recommendation with some being more project-oriented and others being more policy driven. Each recommendation and its related actions are organized based on topical categories such as management, development and infrastructure, and outreach and education. For each recommendation and its subsequent actions, potential participant groups are listed. Participant groups can fulfill several roles, including those who are primarily coordinating activities to those who most often will provide support through time and/or resources to realize the Plan's recommendations. Throughout different phases of the implementation, it is recognized that these roles and groups may change depending upon the expertise that an action may require.

Another component of the Action Plan is implementation timeframe. Some recommendations and related actions could be completed in the next few years, while others may likely take a decade. It is anticipated that physical, policy, and economic influences will impact the feasibility and priorities of the recommendations. Consequently, the Action Plan is intended to respond to opportunities that emerge, issues that arise, and projects that are completed from year to year. Based upon these aspects, the Action Plan should be reviewed and updated on a yearly basis. New actions can be added and assignments of stakeholders refined. Each recommendation can be classified into one of five timeframes:

- Immediate Action –needed to create momentum to move other actions forward;
- High Priority Action – some urgency exists to warrant attention in the short term;
- Medium Priority Action – moderate urgency, can address when resources are available;
- Long-term Priority Action – can be addressed sometime in the future;
- On-Going Action – may start now and continue in the future.

Many of these actions are considered to be building blocks to achieve the recommendations outlined in the conservation strategy and will have to be completed in progression. Items marked as immediate are necessary to create momentum to move the other actions forward. Other actions will be completed as resources and participation become available.

ACTION PLAN		
Recommendations	**Participant Groups**	**Project Initiation Timeline**
1. Management		
A. Identify or form a conservation and management advisory body to implement recommendations of the Conservation and Management Plan (referred to in the recommendations as "river advisory body").	County Governments, municipalities	Immediate
B. Encourage communities to participate in the Act 167 Process	County Governments, municipalities, river advisory body	Immediate
C. Create a Watershed Management Ordinance that promotes watershed conservation principles and techniques	County Governments, municipalities	High Priority
D. Preserve the scenic quality of the Beaver River Corridor including biological diversity areas, natural heritage areas, steep wooded slopes, riparian (streamside) areas, wetlands, islands and other similar characteristics	County Governments, municipalities, conservation organizations, river advisory body	High Priority
E. Preserve and enhance contiguous forests in identified sensitive landscapes by developing a Forest Management Plan for the region	County Governments, municipalities, land owners, district forester, conservation organizations, river advisory body	High Priority
F. Conduct an Agricultural Practices Assessment to identify appropriate farmland conservation and runoff management opportunities	Conservation Districts, County Governments, river advisory body	Long Term
G. Clean up old dock posts from washed out docks	Dock owners, municipalities	Medium Priority
H. Develop, in cooperation with stakeholders, an annual report updating key indicators of watershed health and describe recent progress in preserving and enhancing watersheds, new findings, and study results	River advisory body, Conservation Districts	On-going
I. Gauge and build public support for water conservation and recycling	County Conservation Districts, County Governments, river advisory body, conservation organizations, schools	Medium Priority

J. Conduct a thorough plant and wildlife inventory	Conservation organizations, educational institutions	Medium Priority
K. Clean up illegal dumpsites and litter along the River	Municipalities, conservation or environmental organizations	On-going
L. Study the water quality of the Beaver River watershed	PA DEP, Conservation Districts	High Priority
2. Development and Infrastructure		
A. Encourage redevelopment and in-fill of brownfields (previously developed sites)	County Governments, Municipalities, Development Organizations/Corporations, Authorities, Utilities	On-Going
B. Promote development within existing infrastructure service areas	County Governments, Municipalities, Development Organizations/ Corporations, Authorities, Utilities, river advisory body	On-Going
C. Align new development and future infrastructure expansion to minimize impacts on sensitive natural resource areas	County Governments, Municipalities, Development Organizations/Corporations, Authorities, Utilities	High-Priority
D. Build collaboration between the public and private sectors to update aging and/or under-serviced stormwater management systems	County Governments, Municipalities, Development Organizations/Corporations, Authorities, Utilities, Conservation Districts, river advisory body	On-Going
E. Consider repair and reestablishment of riparian (streamside) vegetation in impacted watersheds	Conservation Districts, river advisory body	Medium Priority
F. Sponsor septic system education for residents of unsewered portions of the Beaver River Corridor	PA DEP, Conservation Districts, river advisory body	High Priority
G. Reduce accumulation of sediment in infrastructure systems by addressing point and nonpoint pollution sources	County Governments, Municipalities, Conservation Districts, river advisory body, PADEP	Medium Priority
H. Address stormwater issues and sedimentation problems within the Beaver River corridor in context of existing and future Act 167 and MS4 plan recommendations	County Governments, Municipalities, PA DEP, Conservation Districts, river advisory body	Medium Priority
I. Remove invasive plant species in public riparian (streamside) areas and in private lands where possible. Continue to monitor riverfront lands for invasive species	Conservation Districts, river advisory body, conservation organizations	On-Going

J. Promote planting of native species in newly developed riverfront areas	Municipalities, Conservation Districts, river advisory body, conservation organizations	High Priority
K. Promote use of pervious pavements, gravel access roads, and other means to encourage stormwater infiltration into the water table	Conservation Districts, Municipalities, new advisory body	Medium Priority
L. Implement riverfront development strategies identified in other studies including the area's Riverfront Development Plan	County Governments, Municipalities, Development Organizations/Corporations	Medium Priority
3. Outreach, Education, and Public Awareness		
A. Pursue funding to support regional-oriented staff for the river advisory body in its coordination and implementation of the Conservation and Management Plan's recommendations	County Governments	High Priority
B. Create a Beaver River Watershed Website related to the projects outlined in the Conservation and Management Plan	County Governments, river advisory body, Conservation Districts	Medium Priority
C. Coordinate conservation and management initiatives with local schools and institutions of higher education to promote research/implementation opportunities including development of primary and secondary school education curricula related to the natural resources of the Beaver River Watershed	Conservation Districts, local non-profits, schools, river advisory body	Medium Priority
D. Work with the railroads, municipalities, and the PA Fish and Boat Commission on designating safe active and passive river access locations	Trail Groups, local non profits, Conservation Districts, PA Fish and Boat Commission, County Governments, Municipalities, river advisory body, railroads	Medium Priority
E. Establish scenic driving tours of the River valley, including stops for train viewing, scenic river views, and historic/cultural attractions	Conservation Districts, local non-profits, historical societies, river advisory body	High Priority
F. Help to coordinate input to, and distribution of, outreach newspapers published by agencies and community groups	Conservation Districts, local non-profits, river advisory body	Medium Priority

G. Encourage and assist local and regional agencies to incorporate interpretive and educational features as part of recreational facilities and other public works projects	Conservation Districts, Municipalities, river advisory body, local non-profits	On-going
H. Promote active stewardship among commercial and residential stakeholders; bring this message/updates to advisory boards, environmental commissions, planning commissions, and other venues for public input to agency decision making	Conservation Districts, local non-profits, conservation organizations, river advisory body	High Priority
I. Sponsor semi-annual hazard and toxic disposal collection programs	Conservation Districts, local non-profits	Medium Priority
J. Establish an Adopt-A-Creek program	Conservation Districts, local non-profits	Long Term

CHAPTER 4: STUDY AREA CHARACTERISTICS

The Beaver River, with its steep slopes and wooded hillsides, is a notable natural and recreational amenity within western Pennsylvania. As part of this environment, there are many characteristics both on the river's banks and on surrounding higher ground that make this area an important consideration for future conservation and watershed management efforts in Beaver and Lawrence Counties.

These characteristics were evaluated as part of this Plan utilizing a series of data organized as part of a Geographic Information System (GIS). Data were compiled from available county mapping files as well as other general resource databases available through the Southwestern Pennsylvania Commission. As part of this compilation, most, but not all, resources were available for both counties. Additional information was compiled from a variety of different sources, including published and personal communication, and noted accordingly within each section.

Study Area

Study Area Location and Size

The composition of the Beaver River Study Area includes more than fifty thousand acres of the lower Beaver River watershed in western Pennsylvania. Specifically, the Study Area includes all lands adjacent to the Beaver River between its formation south of New Castle where the Shenango and Mahoning Rivers merge to its confluence with the Ohio River. The Study Area also includes eighteen individual watersheds. All Study Area lands fall within Beaver and Lawrence Counties and include a significant portion of each, if not all, of the following municipalities:

Beaver County
Beaver Falls City
Big Beaver Borough
Bridgewater Borough
Brighton Township
Chippewa Township
Daugherty Township
Eastvale Borough
Fallston Borough
Homewood Borough
Koppel Borough
New Brighton Borough
North Sewickley Township
Patterson Township
Patterson Heights Borough
Pulaski Township
Rochester Borough
Rochester Township
West Mayfield Borough
White Township

Lawrence County
Little Beaver Township
New Beaver Borough
North Beaver Township
Shenango Township
Taylor Township
Wampum Borough
Wayne Township

A map of the Study Area and descriptions of the subwatersheds appear in Chapter 2.

Socioeconomic Profile

Population

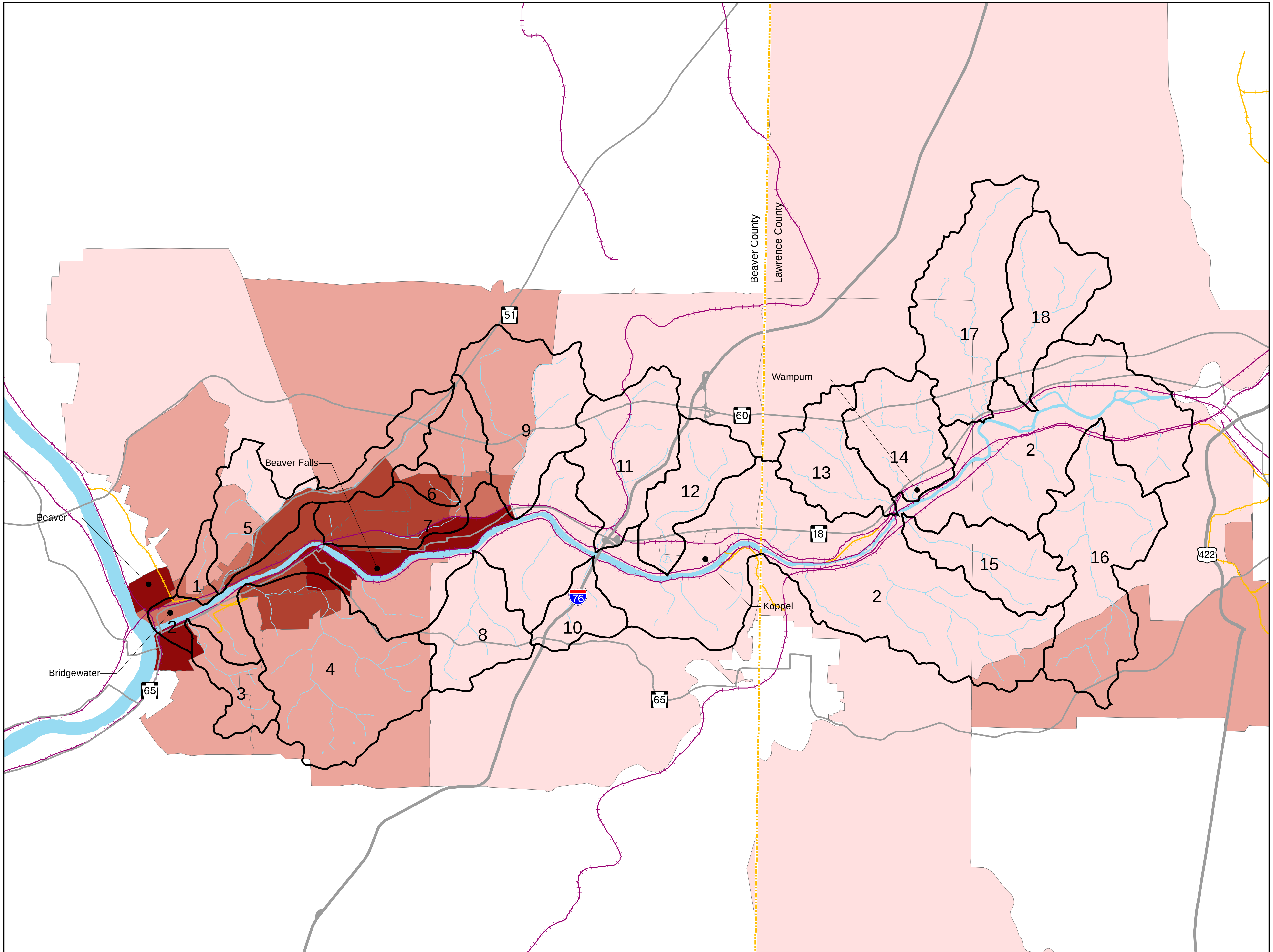
The Beaver River was an ideal location for industry in the early 1900s because of abundant water and rail service. By the late 1900s, however, industry declined in these river communities, accompanied by declines in populations. Some communities have lost a third or more of their population since 1970 as illustrated in Table 4-1. As a result, many communities are seeking new opportunities to lure businesses and residents to the area.

Table 4-1
Municipal Populations

Municipality	2000 Population	1990 Population	1980 Population	1970 Population	% Change 1970 to 2000
Beaver County					
Beaver Borough	4775	5028	5441	6100	-21.7
Beaver Falls City	9920	10687	12525	14375	-31.0
Big Beaver Borough	2186	2298	2815	2739	-20.2
Bridgewater Borough	739	751	879	966	-23.5
Brighton Township	8024	7489	7858	7532	6.5
Chippewa Township	7021	6988	7245	6613	6.2
Daugherty Township	3441	3433	3605	3719	-7.5
Eastvale Borough	293	328	379	421	-30.4
Fallston Borough	307	392	312	571	-46.2
Homewood Borough	147	162	188	212	-30.7
Koppel Borough	856	1024	1146	1312	-34.8
New Brighton Borough	6641	6854	7364	7637	-13.0
North Sewickly Township	6120	6178	6758	6048	1.2
Patterson Heights Borough	670	576	797	777	-13.8
Patterson Township	3197	3074	3288	3442	-7.1
Pulaski Township	1674	1697	1998	2126	-21.3
Rochester Borough	4014	4156	4759	4819	-16.7
Rochester Township	3129	3247	3427	4089	-23.5
West Mayfield Borough	1187	1312	1712	2152	-44.8

White Township	1434	1610	1870	1747	-17.9
Lawrence County					
New Beaver Borough	1677	1736	1426	1885	-11.0
North Beaver Township	4022	3982	3475	4367	-7.9
Shenango Township	7633	7187	7798	7937	-3.8
Taylor Township	1198	1326	1152	1519	-21.1
Wampum Borough	678	666	1189	851	-20.3
Wayne Township	2328	2785	3130	3130	-25.6
Source: http://www.pasdc.hbg.psu.edu/					

The Population Density Map illustrates the current number of people living in each acre of the Study Area. The densest population is generally concentrated within the southern portions of the Beaver River valley. The Boroughs of Beaver Falls, Beaver, and Rochester all indicate a level greater than six people per acre and represent the highest densities in the Study Area. These boroughs are characterized by an environment that has been significantly manipulated to accommodate housing and commercial uses and are typically considered to be “urban.” A population density of less than half a person per acre indicates an area that is primarily rural with minimal infrastructure. These rural areas are generally in the northern portion of the Study Area.



Legend

- Watershed Boundary
- County Boundary
- Rivers/Streams/Lakes
- Highways
- Major Roads
- Active Railroads
- Inactive Railroads

Population Density per Acre

- < 0.5
- 0.5 - 1.5
- 1.5 - 3.0
- 3.0 - 6.0
- > 6.0

Subwatersheds

1. Hamilton Run
2. Beaver River
3. McKinley Run
4. Block House Run
5. Brady Run
6. Grimms Run
7. Walnut Bottom Run
8. Bennett Run
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18. Edwards Run

1,000 Acres
500 Acres
250 Acres

POPULATION DENSITY

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: September 2006
1989.05.16r1

0 1/2 1 MILE

NORTH

Employment

Although agricultural land and forested land comprise 71 percent of the total land area in the study area, the occupational profiles for farming and forestry are minor, accounting for less than one percent for each municipality. The majority of employment opportunities fall within management and professional, service, sales, construction, and production industries. See www.pasdc.psu.edu for the US Census 2000 data on occupational profiles for the Beaver Corridor Municipalities.

Ownership and Control

According to Pennsylvania Common Law, navigable rivers, streams, and lakes are public property. The public has a right to use them for transportation and other purposes without the permission from the owners of streamside properties through which the waters flow. The Beaver River was declared to be navigable by acts of the Pennsylvania legislature during the eighteenth and early nineteenth centuries. Therefore, the Beaver River is public property surrounded mostly by private property (e.g. industrial, residential). Public access areas are needed to allow those who do not own property along the river a means to reach it.

The Commonwealth holds the bed of a navigable waterway in trust for the public in order to protect the public's right to use the waters. The Commonwealth may permit private parties to use the bed of a navigable river for various purposes, for example the dredging of sand and gravel. The Commonwealth also owns the islands in the waterways and can convey them to a private landowner.

Infrastructure

Railroads

Historically, railroads supported the booming industrial communities along the river. While some of those rail lines have been abandoned due to the decline in industry, many are still in use along the river corridor, including rail lines along each bank of the Beaver River. Railroads can be found on the Impoundments Map

Some groups and municipalities are working with the railroad companies to use abandoned rail beds or the space along active rail beds for trails and other recreational uses. This topic is described later in this Chapter.

Highways

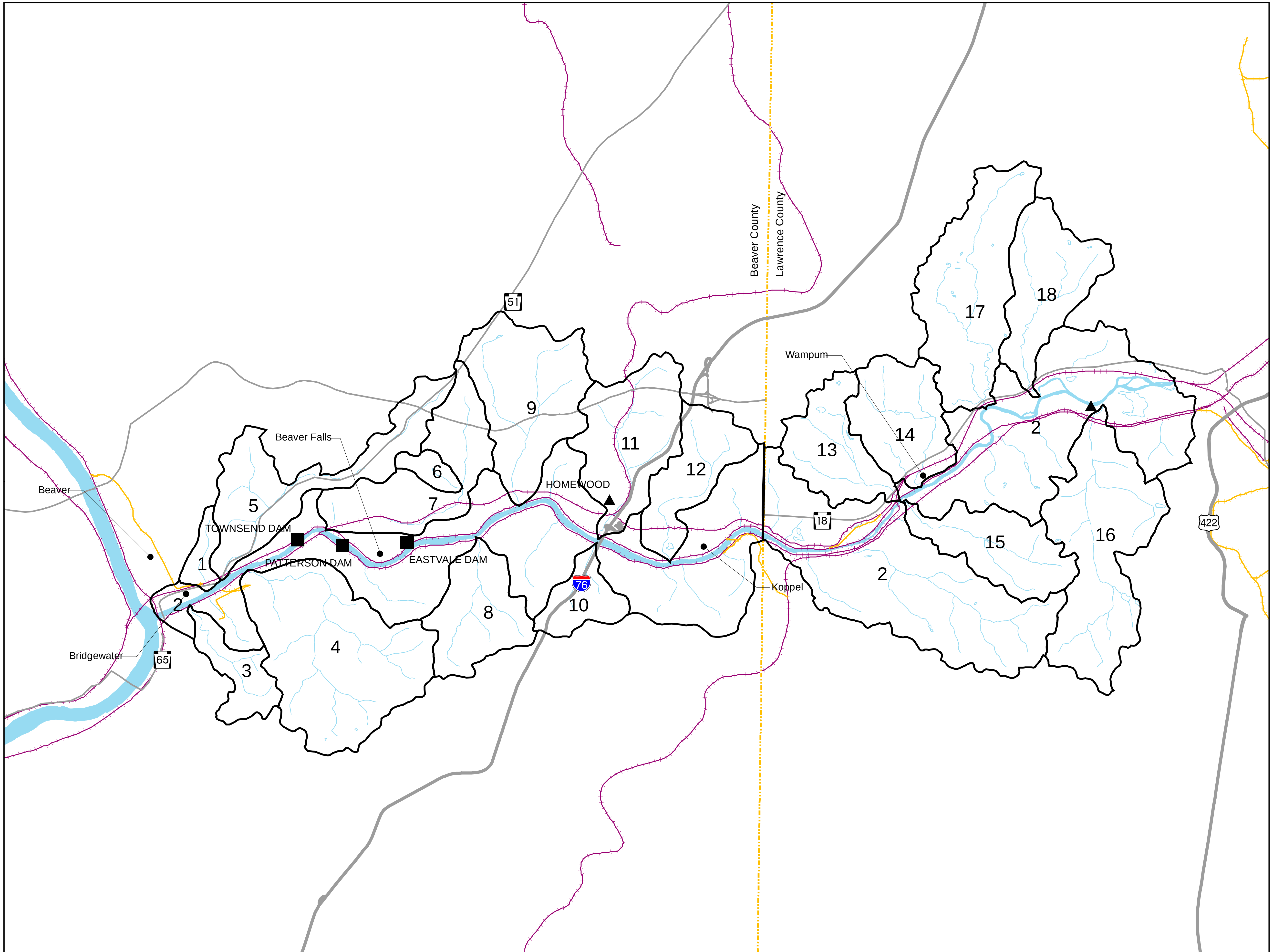
The major highways in the study area include State Routes 18, 51, and 60, and the Pennsylvania Turnpike. Turnpike reconstruction and bridge replacement is occurring west of the Beaver River from milepost 0 at the Ohio border to milepost 10 at the New Castle/Beaver Falls exit. For more information on the construction, visit www.paturnpike.com/constructionprojects/mp0to10/.

The Veteran's Memorial Bridge connecting State Route 51 and Routes 65/18 is slated for construction during 2007-2008. The bridge will use existing bridge piers from the former Sharon Road Bridge that connected Bridgewater and Rochester Township. The bridge will include a pedestrian walkway that may eventually connect with future trails along the east riverbank south to Rochester and along Route 65 to New Brighton in the north.

Dams and Impoundments

The Impoundments Map highlights all human-made reservoirs that currently exist within the Study Area. Most often formed by damming a natural watercourse, an impoundment may be created for a number of practical and/or aesthetic reasons, including hydroelectric power production, water supply, flood control recreation purposes, etc.

Impoundments and the process of damming rivers and streams can have significant environmental impacts. Older dams not equipped with a fish ladder or other means of passing through the structure may disrupt fish migratory patterns. In addition, impoundments often reduce downstream water temperatures, endangering species that are unable to adapt and/or altering species' natural life cycles. The Beaver River is currently dammed in three locations.



Legend

- Watershed Boundary
- County Boundary
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- Active Railroads
- Inactive Railroads
- Hydroelectric Dams
- Other Dams

Subwatersheds

1. Hamilton Run
2. Beaver River
3. McKinley Run
4. Block House Run
5. Brady Run
6. Grimms Run
7. Walnut Bottom Run
8. Bennett Run
9. Wallace Run
10. Thompson Run
11. Clarks Run
12. Stockman Run
13. Wampum Run
14. Eckles Run
15. Snake Run
16. McKee Run
17. Jenkins Run
18. Edwards Run

1,000 Acres
500 Acres
250 Acres

IMPOUNDMENTS

BEAVER RIVER CONSERVATION AND MANAGEMENT PLAN

Prepared for: Pennsylvania Environmental Council
Prepared by: Environmental Planning and Design, LLC

Date: August 2005
1989.05.09

0 1/2 1 MILE

NORTH

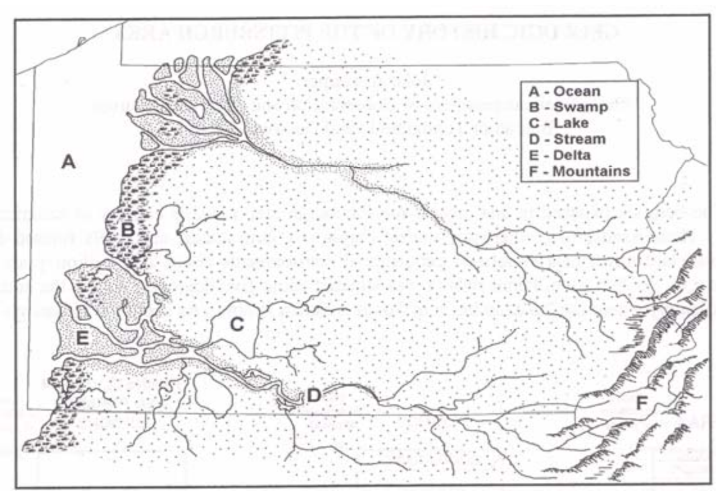
Land Resources

Topography and Geology [1]

Physiographic Provinces, or landforms, are defined by geologists to describe the terrain of large regional landscapes. For instance, landforms differentiate mountainous terrains with steep valleys from plateaus flattened under the oppressive weight of ancient glaciers. The Beaver River watershed spans two sections: the Northwestern Glaciated Plateau and the Pittsburgh Low Plateau.

Formation of the River Valley[2]

About 300 million years ago, western Pennsylvania was the coast of a western inland sea. Two great rivers flowed west across the state, the southernmost one draining at what is now Pittsburgh. Here, a delta formed with deposits of mud, sand, and vegetation, all of which later became shale, sandstone, and coal, respectively. The result is that much of western Pennsylvania now rests on the Main Bituminous coal field.

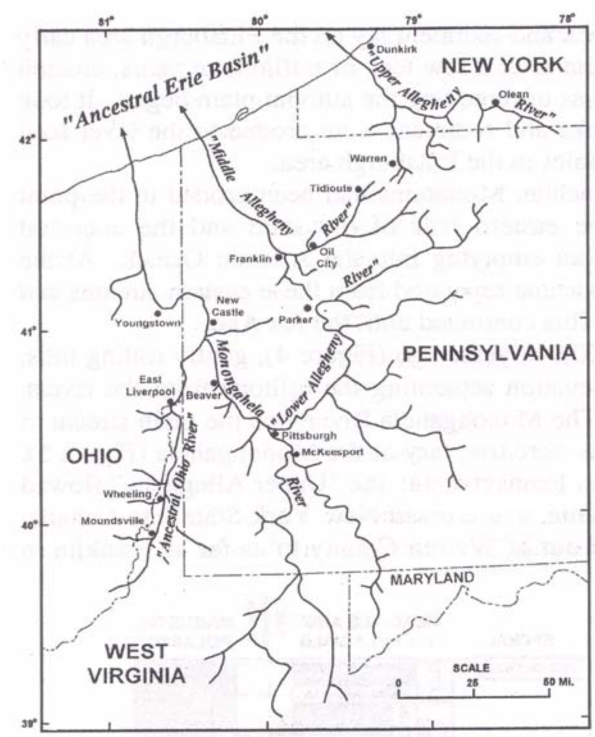


Eventually, millions of years later, the earth's plates began shifting, and the Allegheny Mountains began to form, severing the rivers and forcing new river and stream systems to flow downhill and erode the mountains. Over the centuries, the eroded material was deposited and formed the hills of our landscape that we see today.[3] (See Figure 4-1)

Figure 4-1. *Pennsylvania's geography during the Pennsylvanian Period. Taken from John Harper's Geologic History of the Pittsburgh Area, Department of Conservation and Natural Resources (DCNR).*

Approximately one million years ago, the drainage system of western Pennsylvania was vastly different. At that time, the rivers flowed north toward Canada. The Monongahela River was the dominant river in the system; it flowed along its present day channel, more or less, to Pittsburgh, then along the present channel of the Ohio River to the Beaver River. At that point, the Monongahela River flowed northward along the present day Beaver River Valley, eventually draining into an 'ancestral Erie basin.' The Ohio River was a tributary of the Monongahela, entering it just south of New Castle, Pennsylvania. The Allegheny River was three separate, unrelated rivers with the lower Allegheny River as a tributary of the Monongahela, and the middle and upper Alleghenies flowing directly into the Ancestral Erie Basin. The lower Allegheny River followed the present channel of the Clarion River and flowed south, joining the Monongahela River at Pittsburgh. (See Figure 4-2)

[1] www.dcnr.state.pa.us/topogeo
[2] Harper, John. The Formation of the Allegheny River. Network Notes, December 1996. Volume 1, Issue 1. and April 1997, Volume 1, Issue 2.
[3] Kidney, Walter C. 1982. The Three Rivers. Pittsburgh History and Landmarks Foundation.



During the latter part of the Ice Age, about a half million years ago, the Illinoian glacier moved into northwestern Pennsylvania and blocked the flow of water of the northern flowing rivers. Water flowed over the ridges between the systems and carved out new valleys, took over existing channels, and reversed the flow of the rivers. As a result, the Monongahela River flowed northwest to Pittsburgh where it joined the Allegheny River – now one large river instead of three separate ones. The Allegheny, Monongahela, and the Beaver Rivers became tributaries of the Ohio River, which now drained into the Mississippi River. (See Figure 4-3)

Figure 4-2. *The rivers of western Pennsylvania once flowed north to an ancestral Erie Basin. Taken from John Harper's Geologic History of the Pittsburgh Area, DCNR.*

The retreat of the glaciers provided the river systems with additional water and energy to transport silt, sand, and gravel that had been brought to Pennsylvania by the glacier. This glacial sand and gravel would be extracted many years later as industry along these rivers developed. The land in western Pennsylvania, which had been depressed by the weight of the glaciers, rose after their retreat. Rivers were forced to cut new channels as old river valley floors were now high above the streams.

The next major glaciation, the Wisconsinian, advanced into Pennsylvania 75,000 years ago. This glacial event added silt, sand, and gravel to the Allegheny and Ohio River valleys and caused the Monongahela River and its tributaries to build up their channels with sediments. By the time the Ice Age ended 10,000 years ago, the volume of water and the sediments in the rivers had declined. The rivers cut new, shallow channels in the sand and gravel, ultimately creating the river system seen today.

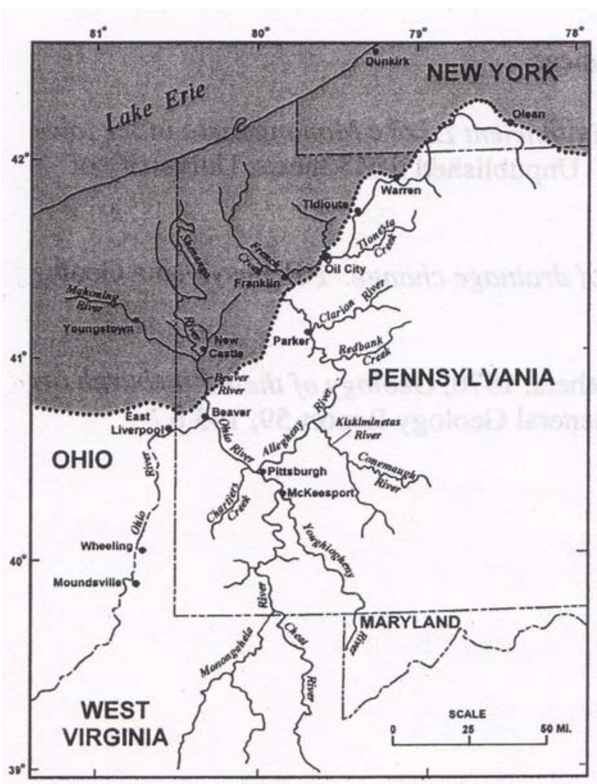


Figure 4-3. *The formation of today's rivers by the southward flow of the glaciers (represented by the shaded area). Taken from John Harper's Geologic History of the Pittsburgh Area, DCNR.*

Geology

Within the Study Area Watershed, there are three types of geologic formations as shown on the Geology Map: sandstone, shale-sandstone (predominately shale) and sandstone-shale (predominately sandstone). Sandstone-shale geologic material characterizes approximately 75 percent of the Study Area, and is therefore the major geologic formation in the Study Area. Sandstone is a sedimentary rock composed mainly of quartz and/or feldspar. Some types of sandstone are resistant to weathering. Sandstone is a common building and paving material. Rock formations that are primarily sandstone usually allow percolation of water and are porous enough to store large quantities of water. This type of material typically filters out pollutants from the land surface.